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Translational control in the spinal cord regulates gene expression and pain hypersensitivity in the chronic phase of neuropathic pain

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eLife Assessment

Using a combination of innovative and robust techniques, this study outlines cell-type-specific translational landscape changes that occur in the spinal cord neurons in the early and late phases of nerve injury. The authors provided **compelling** evidence suggesting an essential role of protein synthesis regulation in the chronic phase of neuropathic pain. Although additional mechanisms contributing to late-phase neuropathic pain beyond altered PV+ neuron excitability remain to be elucidated, this is a **fundamental** and significant study toward a comprehensive understanding of the molecular pathways involved in neuropathic pain.

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Abstract

Sensitization of spinal nociceptive circuits plays a crucial role in neuropathic pain. This sensitization depends on new gene expression that is primarily regulated via transcriptional and translational control mechanisms. The relative roles of these mechanisms in regulating gene expression in the clinically relevant chronic phase of neuropathic pain are not well understood. Here, we show that changes in gene expression in the spinal cord during the chronic phase of neuropathic pain are substantially regulated at the translational level. Downregulating spinal translation at the chronic phase alleviated pain hypersensitivity. Cell-type-specific profiling revealed that spinal inhibitory and excitatory neurons exhibited substantial changes in translation after peripheral nerve injury. Notably, increasing translation selectively in all inhibitory neurons or parvalbumin-positive (PV⁺) interneurons, but not excitatory neurons, promoted mechanical pain hypersensitivity. Furthermore, increasing translation in PV⁺ neurons decreased their

intrinsic excitability and spiking activity. Conversely, reducing translation in spinal PV⁺ neurons prevented the nerve injury-induced decrease in excitability but did not alleviate mechanical hypersensitivity. Together, these findings advance our understanding of translational control mechanisms in the spinal cord during neuropathic pain and highlight their cell-type- and phase-specific contributions to gene expression and pain hypersensitivity.

Introduction

Peripheral nerve injury may result in neuropathic pain, a debilitating and hard-to-treat condition^{1–3}. The development (early) and maintenance (late) phases of neuropathic pain are mediated by structural and functional changes in peripheral and central pain-processing compartments via complex interactions between neuronal and non-neuronal cells^{3–6}. The persistence of these changes relies on *de novo* gene expression, which is tightly regulated, primarily via transcriptional and translational control mechanisms. Whereas previous studies have characterized transcriptional^{7–11} and translational^{12,13} changes in the dorsal root ganglia (DRG) and spinal cord following peripheral nerve injury and have demonstrated their important roles during the early stage of neuropathic pain^{12,14}, the investigations of these mechanisms in the late maintenance phase are lacking.

Studies in neuronal and non-neuronal cells have revealed a poor correlation between the expression levels of distinct mRNAs and the abundance of their corresponding proteins^{15,16}. The regulation of mRNA translation significantly affects the cellular proteome, representing an important mechanism to account for the discordance between mRNA and protein expression^{17,18}. Thus, it is essential to investigate the role of translational control in regulating gene expression and pain hypersensitivity during the clinically relevant late phase of neuropathic pain.

Recent methodological advances enable the investigation of genome-wide transcriptional and translational changes (using ribosome profiling [Ribo-seq]¹⁹), as well as the identification of actively translating mRNAs in specific cell types (using translating ribosome affinity purification [TRAP]²⁰). Here, we employed Ribo-seq and TRAP techniques to study alterations in gene expression in DRG and spinal cord in the early and late phases of neuropathic pain. We found that both transcriptional and translational mechanisms regulate changes in gene expression in the DRG in the early phase (4 days after nerve injury) and the late phase (63 days after nerve injury), as well as in the spinal cord in the early phase. Surprisingly, changes in gene expression in the spinal cord in the late phase of neuropathic pain were regulated more extensively at the translational level. Targeting a key translation initiation factor, eukaryotic translation initiation factor 4E (eIF4E), in the spinal cord provided a long-lasting alleviation of evoked and spontaneous pain during the maintenance phase. Cell-type-specific translational profiling, using metabolic labeling and TRAP, revealed greater nerve injury-induced translational changes in spinal inhibitory neurons than in excitatory neurons. Activating translation in all inhibitory or in parvalbumin-positive (PV⁺) interneurons, but not in excitatory neurons, was sufficient to induce mechanical hypersensitivity. However, while inhibiting translation in PV neurons prevented the nerve injury-induced decrease in PV neuron excitability, it was not sufficient to alleviate mechanical hypersensitivity.

Taken together, this study provides a characterization of translational changes in the early and chronic phases of neuropathic pain and reveals a role for spinal translational control in the maintenance of pain hypersensitivity.

Results

Translational regulation of gene expression in the chronic phase of neuropathic pain

To study changes in gene expression at both transcriptional (transcriptome) and translational (translatome) levels, we employed Ribo-seq on DRG and lumbar spinal cord tissue obtained from mice subjected to an experimental assay of peripheral nerve damage-induced (i.e., neuropathic)

pain, spared nerve injury (SNI, Fig. 1A [↗](#))²¹, or sham surgery. SNI prominently features mechanical hypersensitivity, which develops within 2-4 days of the nerve injury and persists for many months²². L3-L5 DRGs and the ipsilateral dorsal half of the corresponding segment of the lumbar spinal cord (illustrated in a schematic diagram in Fig. 1A [↗](#)) were collected on day 4 (early phase) and day 63 (late phase) post-SNI and processed for Ribo-seq. Ribo-seq allows the identification of mRNA fragments (ribosome footprints [rFPs]) where translating ribosomes are bound. These mRNA fragments thus remain protected from nuclease-mediated RNase degradation, and thereby reveal the number and location of ribosomes on specific transcripts (Fig. 1B [↗](#))¹⁹. Normalization to the corresponding transcript abundance from the parallel mRNA-seq analysis provides a measure of mRNA translation efficiency (TE) on a genome-wide scale. Using this approach, we identified transcriptionally- and translationally-regulated genes in each tissue and time-point (n = 3 biological replicates/condition, 15 mice pooled per replicate). In the DRG, a significant number of transcripts were altered at both transcriptional and translational levels at day 4 post-SNI (Fig. 1C and D [↗](#) show scatter plots; Fig. S1A [↗](#) provides volcano plots for all conditions; full datasets are provided in Supplementary Table 1). At day 63 post-SNI, the number of transcriptionally altered mRNAs in the DRG decreased compared to day 4 (Fig. 1C and D [↗](#); Fig. S1B [↗](#)). In the spinal cord, changes in gene expression were less pronounced and on day 4 post-SNI, a comparable number of differentially transcribed genes and differentially translated genes were identified (Fig. 1D and E [↗](#); Fig. S1A [↗](#)). Surprisingly, on day 63 post-SNI, changes in gene expression in the spinal cord occurred predominantly at the translational but not transcriptional level (Fig. 1D and E [↗](#); Fig. S1B [↗](#)). Gene Ontology (GO) pathway analysis of translationally regulated genes in the spinal cord on day 63 post-SNI revealed alterations in processes related to extracellular matrix (ECM) organization and its interaction with cell surface receptors, cell adhesion, and protein turnover (Fig. 1F [↗](#)). Collectively, these results indicate that both transcriptional and translational mechanisms mediate changes in gene expression in the DRG in both the early and late phases of neuropathic pain and in the spinal cord in the early phase. In the late chronic phase, however, changes in gene expression in the spinal cord are controlled more prominently at the translational level.

Suppression of spinal translation alleviates established pain hypersensitivity

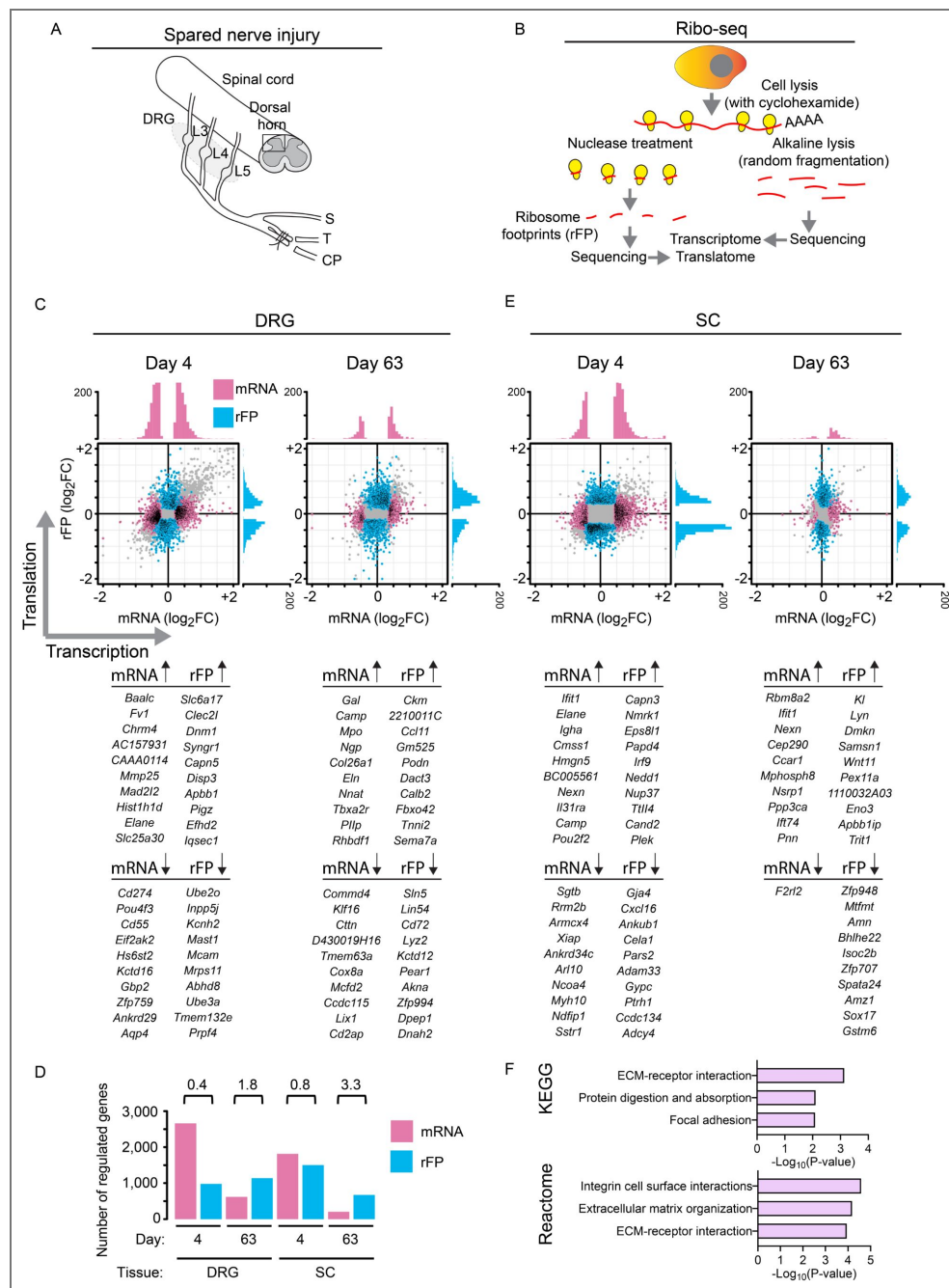


Figure 1. Transcriptional and translational analysis of gene expression using Ribo-seq.

(A) A schematic diagram showing the spared nerve injury (SNI) assay. S: Sural branch, T: Tibial branch and CP: Common peroneal branch. (B) An illustration of the ribosome profiling technique (Ribo-seq). Scatter plot shows ribosomal footprint (rFP) log₂ fold change (FC), reflecting translational changes, as a function of mRNA log₂ fold change for dorsal root ganglia (DRG, C) and spinal cord (SC, E), at day 4 and day 63 post-SNI in female mice. Each dot is a gene. Fold change evaluated between SNI and sham conditions. Color coding indicates modality of differential gene expression control, either at the transcriptional level (mRNA, magenta) or at the translational level (rFP, blue). Under each scatter plot, a list of the top 10 upregulated and downregulated genes (at the mRNA and rFP levels) is shown for each condition. Supplementary Figure 1 shows volcano plots for all conditions. Supplementary Table 1 includes complete datasets (worksheets 1–4), with yellow highlighting category of change in gene expression (translation only, transcription only, stable, opposite change, and homodirectional), grey indicating mRNA log₂FC, and blue indicating rFP log₂FC. (D) Number of genes showing changes at mRNA and rFP levels across independent biological replicates. The rFP/mRNA ratio for each condition is shown above the columns. (F) Pathway analyses of translationally regulated genes in the SC at day 63 post-SNI in the KEGG and Reactome databases.

The important role of spinal translation in regulating changes in gene expression in the chronic phase of neuropathic pain prompted us to test whether targeting translation in the spinal cord can alleviate established pain hypersensitivity. Translation initiation and the activity of cap-binding protein eIF4E, which facilitates the recruitment of ribosomes to the mRNA, are rate-limiting steps in protein synthesis^{23,24}. eIF4E is a key translation initiation factor regulating pain-related plasticity as it integrates information from two signaling pathways that are activated in neuropathic pain to stimulate mRNA translation and promote pain hypersensitivity: the mechanistic target of rapamycin complex 1 (mTORC1), and mitogen-activated protein kinases (MAPKs), such as ERK and p38^{18,25,26} (Fig. 2A). Moreover, previous studies have revealed that a partial reduction in eIF4E expression (e.g. in *Eif4e*^{+/-} mice and in mice treated with eIF4E shRNA) is both well-tolerated and alleviates adverse phenotypes in cancer²⁷ and autism spectrum disorder (ASD)^{28,29} mouse models. To study the role of mRNA translation in chronic pain hypersensitivity, we used an antisense oligonucleotide (ASO) against mouse *Eif4e* to modulate eIF4E expression. To target eIF4E in the central nervous system but not the DRG, we injected eIF4E-ASO via the intracerebroventricular (i.c.v., 100 mg/kg) route³⁰, resulting in a ~38% reduction in eIF4E protein expression in the lumbar spinal cord 2 weeks post-injection (Fig. 2B), without changing eIF4E levels in the DRG (Fig. 2C). We first assessed the effect of eIF4E downregulation on established pain hypersensitivity. Injection of eIF4E-ASO at week 6 post-SNI alleviated mechanical pain hypersensitivity in the von Frey test two weeks later (at week 8 post-SNI; the experimental time course is provided in Fig. 2D, von Frey data in Fig. 2E). Reduced hypersensitivity persisted for 4 additional weeks (up to week 12 post-SNI), demonstrating a long-lasting effect following a single eIF4E-ASO administration. eIF4E-ASO also attenuated spontaneous pain, as assessed using the Mouse Grimace Scale (MGS) on week 8 post-SNI (Fig. 2F).

We then studied the effect of downregulating eIF4E during the early development phase of neuropathic pain by administering eIF4E-ASO and control-ASO ten days before the peripheral nerve injury. Surprisingly, we found no alleviation of mechanical hypersensitivity at day 4 post-SNI in eIF4E-ASO-injected mice (the experimental time course is provided in Fig. 2G, von Frey data in Fig. 2H). However, testing at later time-points showed that mice injected with eIF4E-ASO exhibited reduced mechanical pain hypersensitivity at week 2 after the nerve injury (Fig. 2H) and the effect became more pronounced at weeks 4 and 8. The MGS was also reduced in eIF4E-ASO-injected mice at week 4 and 8 post-SNI (Fig. 2I). To control for potential non-specific effects of eIF4E-ASO following i.c.v. administration, we performed rotarod and open field tests, which revealed no differences in locomotor function between mice injected with eIF4E-ASO and control ASO (Fig. 2J and K). Together, these results indicate that downregulation of eIF4E in the spinal cord using ASO alleviates pain hypersensitivity in the late but not the acute stages of neuropathic pain.

Cell-type-specific translational profiling after peripheral nerve injury

Ribo-seq provided a comprehensive characterization of translational landscape in DRG and spinal cord tissues during the early and late phase of neuropathic pain. However, this approach does not allow the measurement of gene expression in distinct neuronal subtypes and cannot distinguish between neuronal and non-neuronal cells. To assess protein synthesis in specific neuronal subpopulations, we used fluorescence noncanonical amino acid tagging (FUNCAT)^{31,32}, focusing on two major neuronal subtypes in the spinal cord, excitatory and inhibitory neurons. In FUNCAT, mice are injected with a noncanonical amino acid, azidohomoalanine (AHA), which is charged onto methionine tRNA and incorporated into newly synthesized proteins (Fig. 3A). Visualization of AHA incorporation using click chemistry and fluorescent labeling provides a measure of *de novo* general protein synthesis in spinal cord sections. The specificity of this approach was validated using a protein synthesis inhibitor, anisomycin, that blocked AHA incorporation in the spinal cord (Fig. 3B). FUNCAT analysis showed that AHA incorporation increased at day 4 and day 60 post-SNI in Pax2⁺ inhibitory neurons (day 4: Fig. 3C; day 60: Fig. 3D), whereas no statistically significant changes were found in excitatory neurons (NeuN⁺ and Pax2⁻).

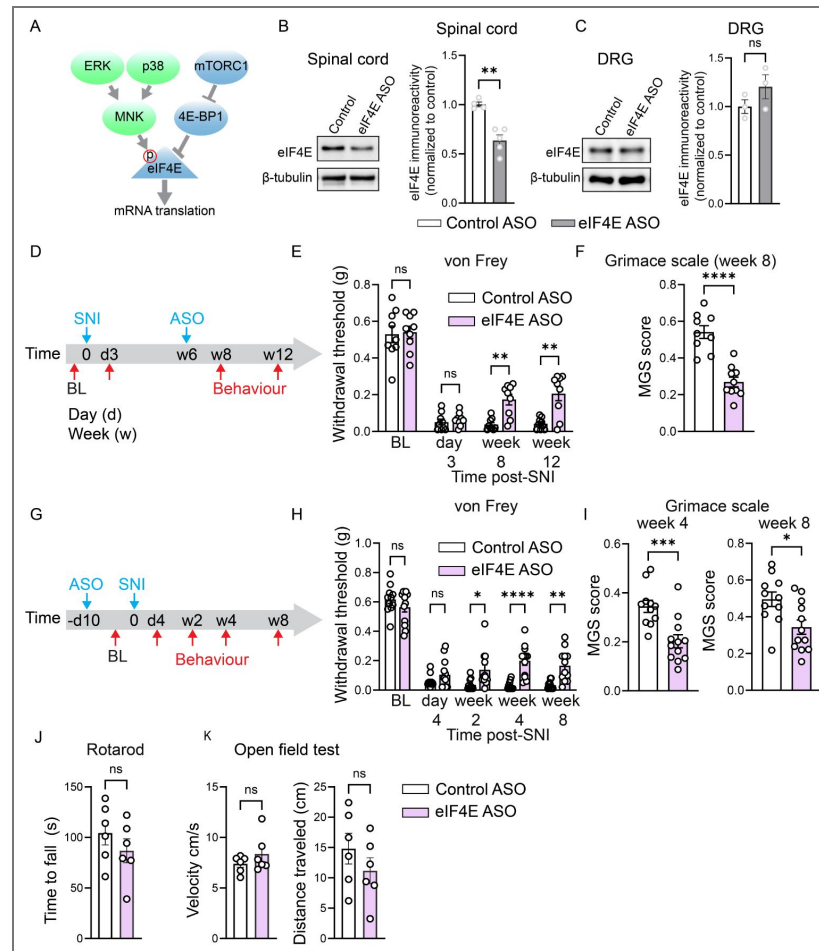


Figure 2. Targeting spinal translation alleviates pain hypersensitivity at the late stage after peripheral nerve injury.

(A) A schematic showing the regulation of eIF4E via mTORC1/4E-BP1 and MAPKs/MNK pathways. eIF4E ASO (i.c.v) reduces eIF4E protein levels in the spinal cord (B) but not DRG (C) two weeks after administration ($n = 4/\text{group}$). (D) Time course of ASO (eIF4E and control) administration after SNI. The effect of ASO on von Frey (50% withdrawal threshold: E, $n = 9/\text{group}$) and Mouse Grimace Scale (MGS) (F, $n = 9/10$ mice per group). (G) Time course of ASO administration before SNI and its effect on the von Frey (50% withdrawal threshold: H, $n = 11/12$ mice per group) and MGS (I, $n = 11/12$ mice per group) tests. An unpaired two-tailed t -test was used in B, C, F, and I. Two-way ANOVA followed by Tukey's post-hoc comparison was used in E and H. Each data point represents an individual animal. A comparable number of male and female mice was used in all experiments. Data are plotted as mean $\pm$ s.e.m. $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, $****p < 0.0001$, ns – not significant.

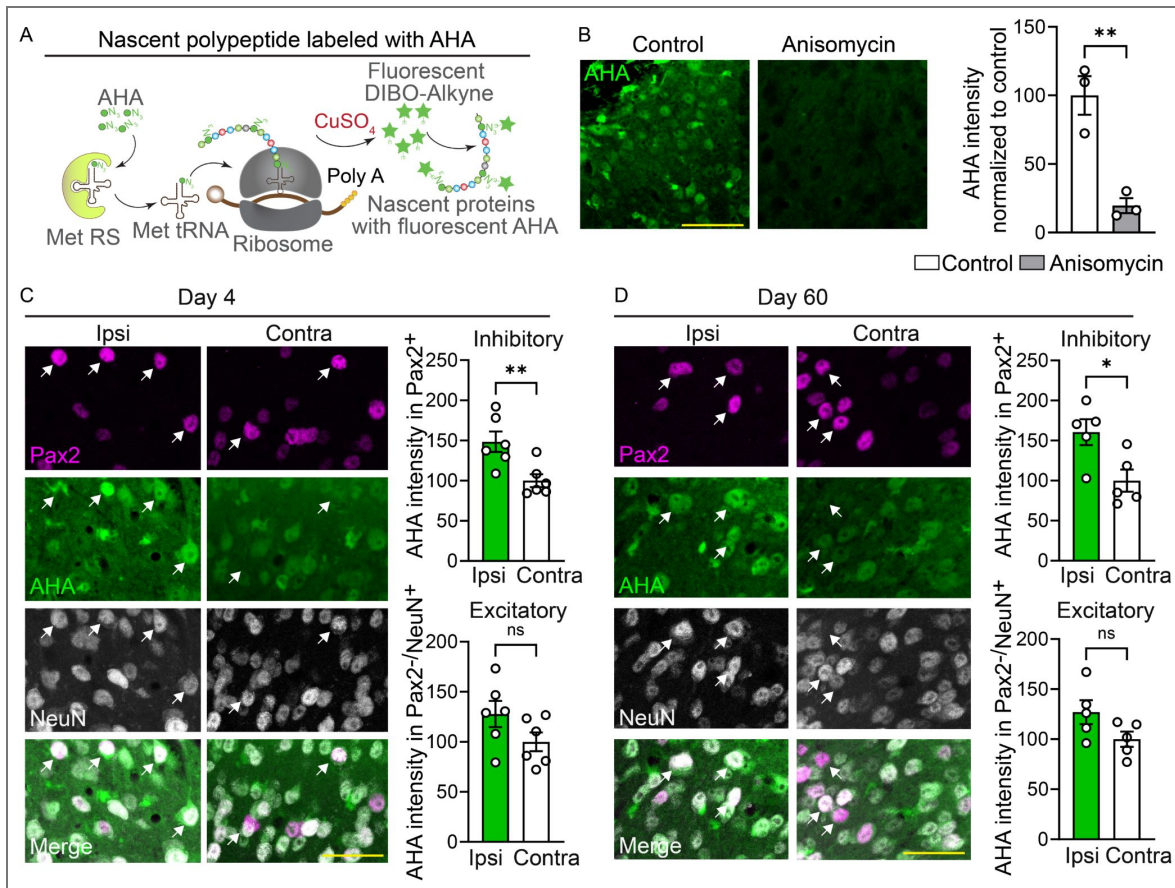


Figure 3. Assessment of protein synthesis using metabolic labeling.

(A) Illustration of protein synthesis assessment using FUNCAT. Anisomycin (100 mg/kg, i.p. injection 1 h before AHA injection) treatment blocked AHA incorporation ($n = 3$ female mice per group, normalized to the control group), demonstrating the validity of the approach. AHA signal in the superficial spinal cord (laminae I-III, defined based on NeuN staining) was quantified in inhibitory neurons (Pax2⁺, examples marked by white arrow) and excitatory neurons (Pax2⁺/NeuN⁺) at day 4 (C, $n = 6$ female mice per group) and day 60 (D, $n = 5$ female mice per group) post-SNI. AHA signal intensity (integrated density on maximum-intensity projection images) in the soma of inhibitory and excitatory neurons were averaged across 25 cells/mouse to obtain a single value for each mouse (see Methods for details of the analysis). Scale bars: 50 μm for B and 30 μm for C, D. An unpaired two-tailed t -test was used. Each data point represents an individual animal. Data are plotted as mean $\pm$ s.e.m. * $p < 0.05$, ** $p < 0.01$, ns – not significant.

Next, we employed the TRAP approach to identify specific mRNAs that are actively translated in excitatory and inhibitory neurons. In TRAP, the eGFP-tagged ribosomal protein, L10a, is expressed in a genetically defined cellular population (via a specific gene promoter), followed by immunoprecipitation of tagged ribosomes with an anti-eGFP antibody and the sequencing of ribosome-bound mRNAs (Fig. 4A). We performed TRAP analysis on two major sub-populations of neurons: a subset of excitatory neurons, defined by *Tac1* (using *L10a*-eGFP; *Tac1*^{Cre} mice); and inhibitory neurons, defined by *GAD2* (using *L10a*-eGFP; *Gad2*^{Cre} mice). *Tac1* is expressed in a subset of excitatory interneurons and projection neurons in the spinal cord that play important roles in processing nociceptive information, as well as driving spinal plasticity and chronic pain-related behaviors^{33,34}. *GAD2*⁺ neurons encompass numerous sub-populations of spinal cord inhibitory neurons that are critical for the development and maintenance of neuropathic pain^{35,36}. To this end, *L10a*-eGFP; *Tac1*^{Cre} and *L10a*-eGFP; *Gad2*^{Cre} mice were subjected to SNI or sham surgery (bilaterally), and lumbar dorsal spinal cord tissue was collected at day 4 and 60 after the nerve injury. mRNAs isolated from the immunoprecipitated (IP) and input (IN) samples were sequenced. Expression levels (IP/IN) of excitatory (*Slc17a7*, *Tac1*, *Cck*, *Nts*) and inhibitory (*Slc32a1*, *Gad2*, *Pax2*, *Pvalb*) neuronal markers, as well as markers of non-neuronal cells (e.g., *Aldh1l1*, *Gfap*, *Tmem119*, *Cx3cr1*), are shown in Figure 4B for *L10a*-eGFP; *Gad2*^{Cre} mice and in Figure 4C for *L10a*-eGFP; *Tac1*^{Cre} mice, demonstrating the specificity of the approach. Changes in ribosome occupancy were found in 126 mRNAs in the early phase and 223 mRNAs in the late phase in *GAD2*⁺ neurons, and 118 mRNAs in the early phase and 161 in the late phase in *Tac1*⁺ neurons (IP: Fig. 4D-H, IP/IN datasets are provided in Supplementary Table 1), suggesting higher translational changes at day 60 post-SNI in *GAD2*⁺ inhibitory neurons compared to *Tac1*⁺ excitatory neurons. Together, these results establish translational changes in *GAD2*⁺ and *Tac1*⁺ neurons in the early and late phases of neuropathic pain.

Upregulating eIF4E-dependent translation in inhibitory neurons promotes pain hypersensitivity

The pronounced upregulation of mRNA translation in inhibitory and excitatory neurons after peripheral nerve injury prompted us to study its functional role in mediating pain hypersensitivity. mTORC1, a master regulator of mRNA translation, stimulates protein synthesis via phosphorylation of the translational repressor eIF4E-binding proteins (4E-BPs), triggering their dissociation from eIF4E to allow cap-dependent translation initiation (Fig. 5A). Accordingly, deletion of 4E-BPs, which mimics the activation of the mTORC1-eIF4E axis, stimulates translation. There are three isoforms of 4E-BP (4E-BP1, 4E-BP2, and 4E-BP3), which exhibit similar functions but have different tissue distribution³⁷. 4E-BP1 is the main isoform in the pain pathway, as 4E-BP1, but not 4E-BP2 whole-body knockout mice show mechanical pain hypersensitivity³⁷, while 4E-BP3 expression is very low in the nervous system. To increase translation selectively in inhibitory or excitatory neurons, we generated mice lacking 4E-BP1 in each cell type (confirmation of 4E-BP1 downregulation is shown in Fig. S2A-D). Deletion of 4E-BP1 in inhibitory neurons (*Eif4ebp1*^{fl/fl}; *Gad2*^{Cre}) induced mechanical hypersensitivity without affecting heat sensitivity (Fig. 5B). A subpopulation of inhibitory neurons, PV⁺ interneurons, specifically gate mechanical allodynia^{38–40}. Peripheral nerve injury induces substantial plasticity in PV neurons, resulting in a decrease in their intrinsic excitability and spiking activity, and the consequent disinhibition of postsynaptic PKC γ interneurons and engagement of myelinated primary afferents in spinal nociceptive circuits^{38–40}. Deletion of 4E-BP1 in PV neurons (*Eif4ebp1*^{fl/fl}; *Pv*^{Cre}) induced robust mechanical hypersensitivity, similar to *Eif4ebp1*^{fl/fl}; *Gad2*^{Cre} mice (Fig. 5C; no change was found in heat sensitivity). Recording from lumbar spinal cord slices showed that PV neurons lacking 4E-BP1 exhibit reduced excitability, as evident by a decreased firing rate in response to a depolarization pulse (Fig. 5D) and elevated rheobase compared to PV neurons from control mice (Fig. 5E). No change was observed in membrane capacitance (Fig. 5F), resting membrane potential (Fig. 5G), and input resistance (Fig. 5H). To study the role of translation in peripheral nerve injury-induced plasticity in spinal PV neurons, we selectively downregulated eIF4E in PV neurons in the lumbar spinal cord before SNI. To this end, an adeno-associated virus (AAV)-expressing shRNAmir against eIF4E (AAV9-CAG-DIO-eGFP-eIF4E-shRNAmir) was injected

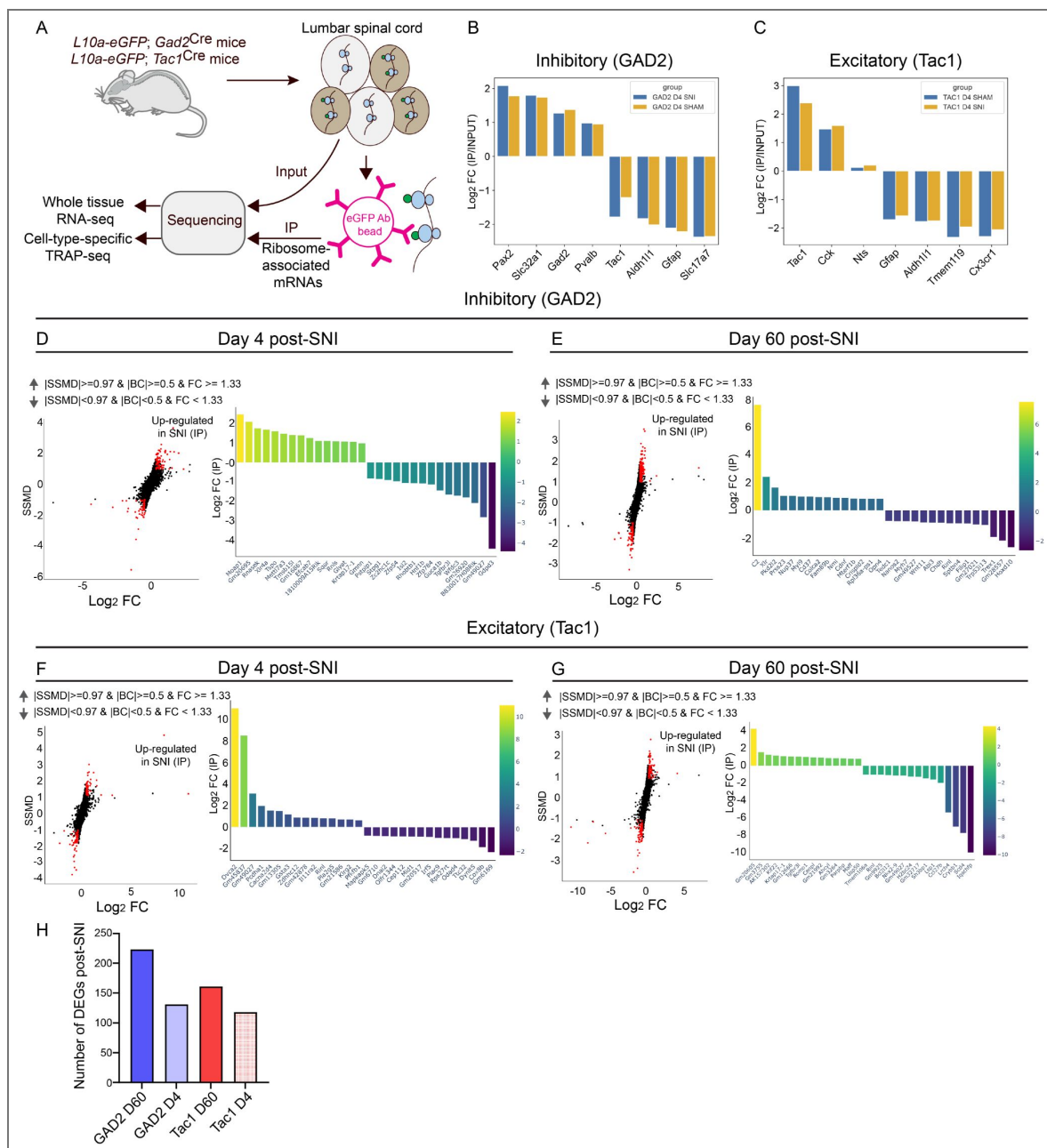


Figure 4. Cell-type-specific profiling of spinal gene expression after peripheral nerve injury.

(A) A schematic illustrating the TRAP approach to assess gene expression in specific cell types. Confirmation of the specificity of IP fractions for inhibitory neurons in the *L10a-eGFP; Gad2^{Cre}* mouse line (B) and for excitatory neurons in the *L10a-eGFP; Tac1^{Cre}* mouse line (C). Experiments were performed in female mice. Dual flashlight plots (left) show the strictly standardized mean difference (SSMD) versus $\text{Log}_2 \text{FC}$ for genes in IP samples and panels on the right show the top 15 upregulated and downregulated genes for inhibitory neurons at day 4 (D) and 60 (E), and Tac1⁺ excitatory neurons at day 4 (F) and 60 (G) post-SNI. Positive $\text{Log}_2 \text{FC}$ indicates increased expression in SNI compared to sham mice. Parameters for defining data as upregulated in SNI are indicated at the top. Supplementary Table 1 includes complete datasets (worksheets 5–8), with yellow highlighting $\text{Log}_2 \text{FC}$ values in IP samples and orange highlighting $\text{Log}_2 \text{FC}$ values in IN samples. (H) The number of altered genes in each condition (GAD2 D60: SNI versus sham day 60 in GAD2⁺ neurons; GAD2 D4: SNI versus sham day 4 in GAD2⁺ neurons; Tac1 D60: SNI versus sham day 60 in Tac1⁺ neurons; and Tac1 D4: SNI versus sham day 4 in Tac1⁺ neurons).

into the lumbar dorsal horn parenchyma of Pv^{Cre} mice 14 days before the SNI (Fig. 5I [↗](#) shows experimental design, Fig. S2E [↗](#) shows confirmation of reduced eIF4E levels). Recordings from PV neurons in spinal cord slices revealed that downregulation of eIF4E in PV neurons prevented the SNI-induced decrease in intrinsic excitability (Fig. 5J [↗](#)). Whereas PV neurons from control mice (Pv^{Cre} mice injected with AAV9-CAG-DIO-eGFP-eIF4E-scrambled) exhibited reduced spiking activity and increased rheobase 4 weeks post-SNI compared to sham animals, PV neurons with reduced eIF4E showed no change in their excitability after nerve injury (Fig. 5J [↗](#): firing frequency; Fig. 5K [↗](#): rheobase). No change was found in membrane capacitance (Fig. 5L [↗](#)), resting membrane potential (Fig. 5M [↗](#)), and input resistance (Fig. 5N [↗](#)).

We next investigated the effect of downregulating eIF4E-dependent translation in PV neurons on SNI-induced pain hypersensitivity using two complementary approaches. We first downregulated eIF4E by injecting AAV9-CAG-DIO-eGFP-eIF4E-shRNAmir into the lumbar dorsal horn parenchyma of Pv^{Cre} mice 14 days prior to SNI. Unexpectedly, this manipulation did not attenuate mechanical hypersensitivity compared with mice injected with AAV9-CAG-DIO-eGFP-scrambled shRNA (Fig. 6A [↗](#)). To confirm this finding, we used an alternative genetic approach. Specifically, we generated mice expressing a mutated, non-phosphorylatable form of 4E-BP1 (harboring threonine-to-alanine mutations at two key mTORC1 phosphorylation sites, amino acids 37 and 46; referred to as Tg-4EBP1mt)⁴¹ selectively in PV neurons (Tg-4EBP1mt; Pv^{Cre}). In these mice, the non-phosphorylatable 4E-BP1 binds to and inhibits eIF4E activity specifically in PV neurons. Consistent with the results of the AAV-eIF4E-shRNAmir-mediated knockdown experiment, we observed no alleviation of mechanical hypersensitivity in Tg-4EBP1mt; Pv^{Cre} mice compared to control Tg-4EBP1mt animals (Fig. 6B [↗](#)). Together, these results indicate that inhibiting eIF4E-dependent translation in PV neurons is sufficient to prevent the SNI-induced decrease in their intrinsic excitability, but is not sufficient to alleviate SNI-induced mechanical hypersensitivity.

Finally, we generated mice lacking 4E-BP1 in a broad population of excitatory $Vglut2^+$ neurons (*Eif4ebp1*^{fl/fl};*Vglut2*^{Cre}) (Fig. 6C [↗](#)), as well as in a subpopulation of excitatory neurons defined by *Tac1* (*Eif4ebp1*^{fl/fl};*Tac1*^{Cre}) (Fig. 6D [↗](#)). These two mouse lines exhibited no changes in mechanical or heat withdrawal thresholds, suggesting that increasing eIF4E-dependent translation in excitatory neurons does not confer pain hypersensitivity.

Discussion

Previous studies in animal models of neuropathic pain have largely focused on changes in the transcriptome and were mostly limited to early time points after peripheral nerve injury. The growing realization of the important role of translational control in neuronal plasticity and pain sensitization, as well as the uncovering of distinct mechanisms underlying the development and maintenance phases of neuropathic pain^{1,18,42,43}, prompted us to study gene expression at both transcriptional and translational levels during early and late time points after peripheral nerve injury. Unexpectedly, we discovered that gene expression during the maintenance phase in the spinal cord is substantially regulated at the level of mRNA translation. Furthermore, we demonstrated that downregulation of the key translation initiation factor eIF4E in the spinal cord, using cell-type-non-specific ASO, leads to long-lasting alleviation of established pain hypersensitivity.

In the late phase of neuropathic pain, we found both transcriptional and translational changes in gene expression in DRG but greater translational changes in the spinal cord than transcriptional changes. Alterations in the DRG transcriptome are consistent with previous analyses in animal models^{44–46} and human DRG tissue from individuals with neuropathic pain, which revealed substantial transcriptional changes^{8,9} accompanied by neuronal hyperexcitability⁹. Gene expression datasets from human neuropathic spinal cord tissue are not yet available due to paucity of spinal cord samples from individuals with neuropathic pain.

Altered translation at late time points after nerve injury might be linked to maladaptive spinal plasticity. eIF4E downregulation in the spinal cord alleviated pain hypersensitivity at the late, but not early, time point. Since translation is the predominant gene expression mechanism in the spinal cord at the late stage, it is conceivable that downregulation of eIF4E normalizes the

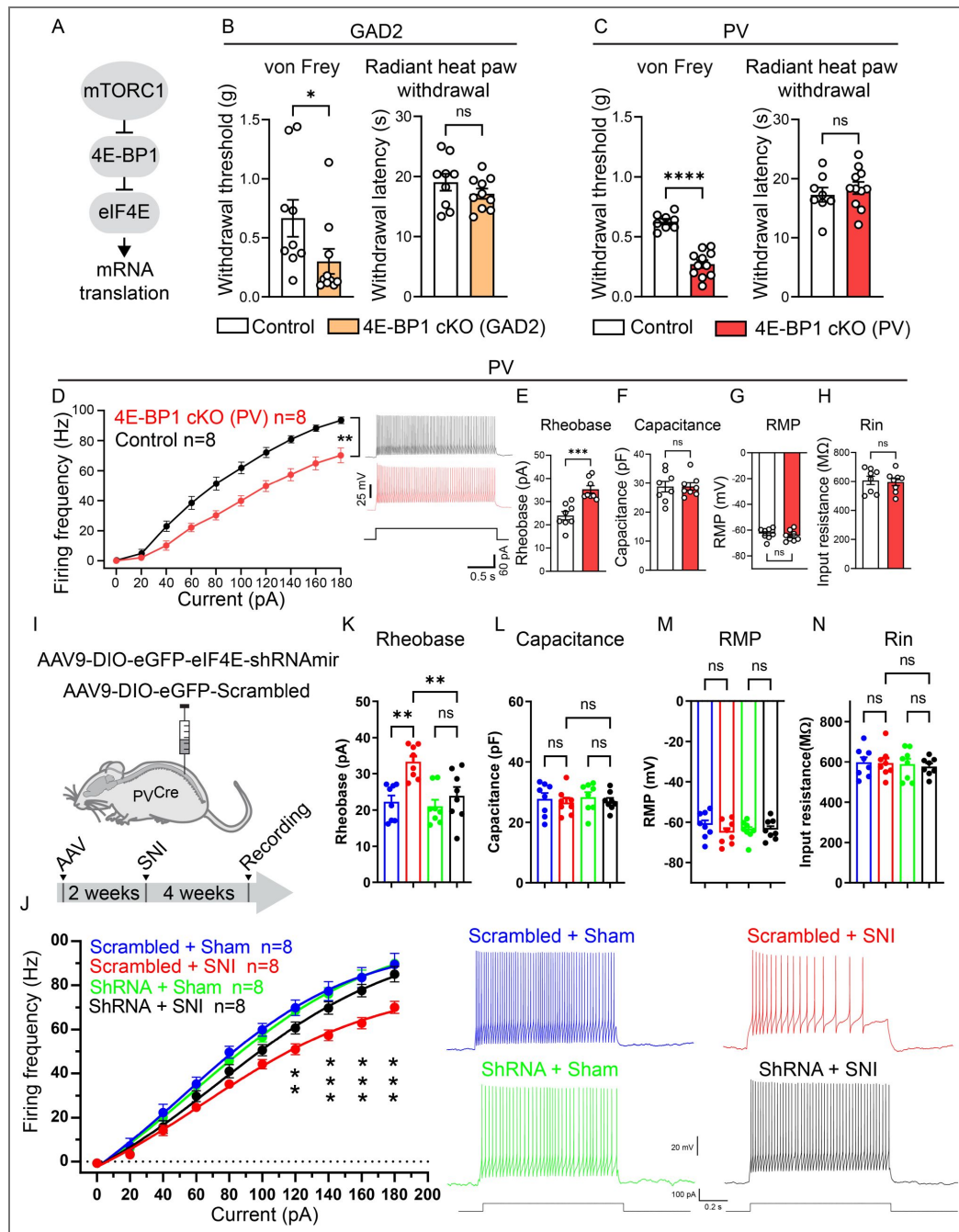


Figure 5. Activation of 4E-BP1-dependent translation in inhibitory neurons promotes mechanical hypersensitivity and contributes to reduced intrinsic excitability of PV neurons.

(A) A schematic of mTORC1 pathway. Deletion of 4E-BP1 in GAD2 (B, 4E-BP1 cKO: *Eif4ebp1*^{fl/fl}; *Gad2*^{Cre}, Control: *Gad2*^{Cre}, n = 9/10) and PV (C, 4E-BP1 cKO: *Eif4ebp1*^{fl/fl}; *Pv*^{Cre}, Control: *Pv*^{Cre}, n = 8/11) neurons induces mechanical (50% withdrawal threshold) but not heat hypersensitivity. A comparable number of male and female mice was used in B and C. Recording from PV neurons in spinal cord slices (identified by the expression of L10a-eGFP) shows that the deletion of 4E-BP1 in PV neurons (4E-BP1 cKO: *Eif4ebp1*^{fl/fl}; L10a-eGFP: *Pv*^{Cre}, Control: L10a-eGFP: *Pv*^{Cre}, n = 8/8 female mice) induces a decrease in firing frequency (D) and an increase in rheobase (E). No change in membrane capacitance (F), resting membrane potential (RMP, G), and input resistance (Rin, H) were found. AAVs (AAV-CAG-DIO-eGFP-eIF4E-shRNA or AAV-CAG-DIO-eGFP-scrambled-shRNA) were injected into the parenchyma of the dorsal horn of *Pv*^{Cre} female mice (illustration and time course are shown in I, n = 8/group), preventing the SNI-induced decrease in PV neuron firing frequency (J) and elevation of rheobase (K). No changes were found in capacitance (L), RMP (M), and Rin (N). An unpaired two-tailed *t*-test was used in B, C, E-H. Two-way ANOVA followed by Tukey's post-hoc comparison was used in J-N. Each data point represents an individual animal. Data are plotted as mean ± s.e.m. **p* < 0.05, ***p* < 0.01, ****p* < 0.001, ns – not significant.

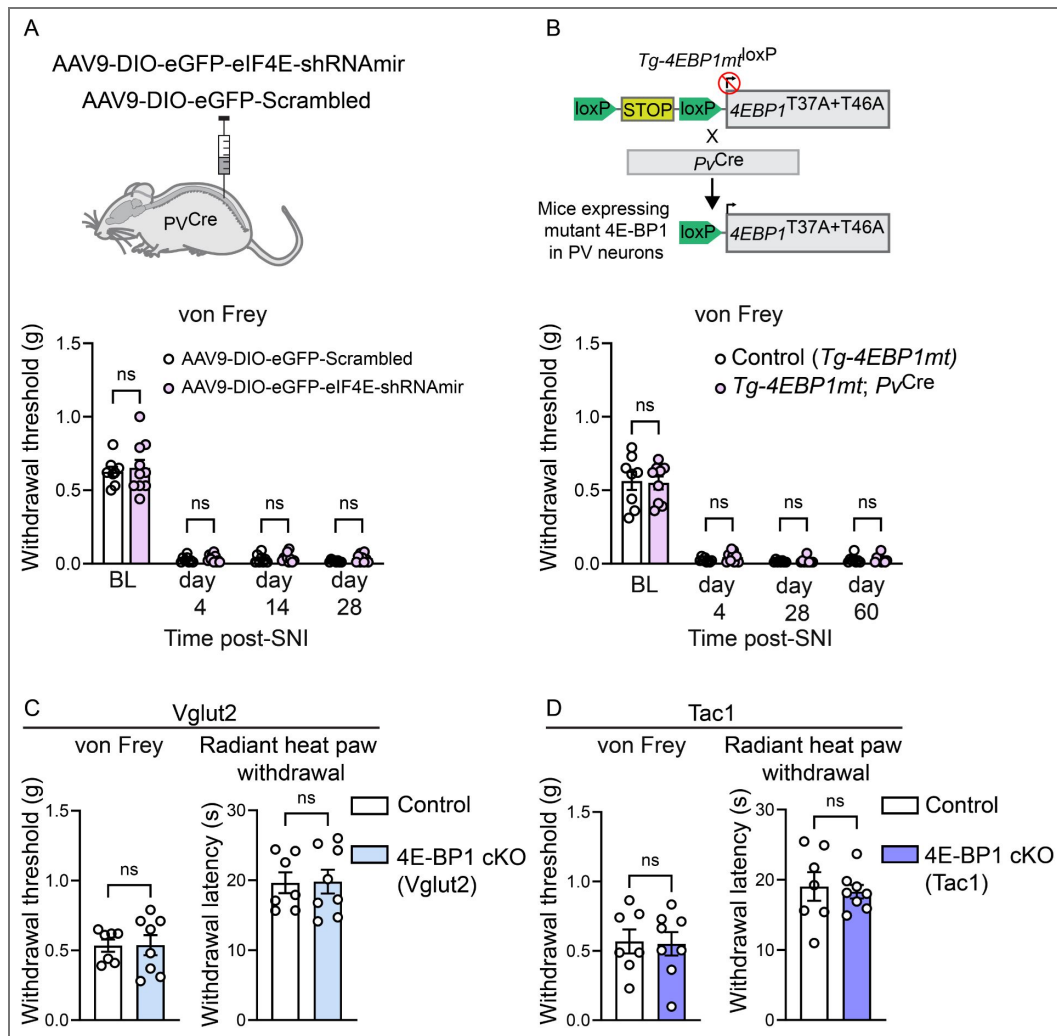


Figure 6. Effects of modulating eIF4E-dependent translation on pain hypersensitivity.

(A) eIF4E was downregulated in PV neurons by intraspinal injection of AAV-CAG-DIO-eGFP-eIF4E-shRNAir (and control AAV-CAG-DIO-eGFP-scrambled-shRNAir) into the parenchyma of the dorsal horn of *PV^{Cre}* mice two weeks before SNI ($n = 10/8$). (B) Mice expressing a mutated non-phosphorylatable 4E-BP1 in PV neurons (*4EBP1^{mt};PV^{Cre}*) and their controls (*Tg-4EBP1^{mt}*) were subjected to SNI ($n = 9/8$). No reduction in SNI-induced mechanical hypersensitivity (von Frey, 50% withdrawal threshold) was observed in A or B. No changes were observed in mechanical (von Frey, 50% withdrawal threshold) and heat (radiant heat paw-withdrawal) thresholds in mice lacking 4E-BP1 in Vglut2 neurons (C, 4E-BP1 cKO: *Eif4ebp1^{fl/fl};Vglut2^{Cre}*, Control: *Vglut2^{Cre}*) or Tac1 neurons (D, 4E-BP1 cKO: *Eif4ebp1^{fl/fl};Tac1^{Cre}*, Control: *Tac1^{Cre}*, $n = 7/8$ mice). A comparable number of male and female mice was used in A-D. An unpaired two-tailed *t*-test. Data are plotted as mean $\pm$ s.e.m. ns – not significant.

translational landscape, thus correcting maladaptive plasticity underlying spinal hyperexcitability. In the early stage after nerve injury, modifications of existing proteins (e.g. via phosphorylation) and transcriptional changes play significant roles^{1,2}, rendering suppression of translation less efficient.

In the maintenance phase of neuropathic pain, we observed greater translational changes, using FUNCAT and TRAP, in spinal inhibitory neurons compared to excitatory neurons. Moreover, enhancing 4E-BP1-dependent translation in GAD2⁺ inhibitory and PV⁺ neurons, but not excitatory neurons, induced mechanical hypersensitivity. Increasing translation was also sufficient to decrease the excitability of spinal PV interneurons, whereas suppressing translation in PV neurons prevented SNI-induced reduction in their excitability. Surprisingly, inhibiting eIF4E-dependent translation in PV neurons using two different approaches (eIF4E downregulation and expression of non-phosphorylatable 4E-BP1) did not alleviate SNI-induced mechanical hypersensitivity. There are several potential explanations for these results, including: (1) the presence of other mechanisms in PV neurons (e.g., a reduction in PV neuron synaptic output post-SNI mediated by retinoic acid receptor RARα⁴⁰) that are translation-independent; (2) the insufficiency of correcting reduced PV neuron excitability to alleviate hypersensitivity; and (3) an essential role for mRNA translation in other neuronal and/or non-neuronal cell types in neuropathic pain. Indeed, the alleviation of neuropathic pain hypersensitivity by eIF4E-ASO might be mediated by effects on multiple neuronal (e.g., excitatory and inhibitory neurons) or non-neuronal (e.g., microglia, astrocytes, immune cells) cell types, whereas reducing eIF4E-dependent translation solely in PV neurons is not sufficient to reverse hypersensitivity.

Spinal disinhibition, induced by peripheral nerve injury, plays a key role in central sensitization. Numerous neuronal and non-neuronal mechanisms contribute to this disinhibition, including: K⁺-Cl⁻ cotransporter (KCC2) downregulation causing elevation of intracellular chloride and the resulting weakening of inhibitory neurotransmission^{47,48}, preferential removal of inhibitory synapses by microglia^{49,50}, and the modulation of the extracellular matrix⁵¹. In addition, peripheral nerve injury induces substantial plasticity in spinal PV neurons, leading to the reduction in their synaptic output as well as intrinsic excitability, thereby resulting in the engagement of myelinated low-threshold mechanoreceptive (LTMR) afferents in spinal nociceptive lamina I circuits^{38–40}. Our data demonstrate that the reduction in PV neuron intrinsic excitability and spiking activity are mediated, at least partially, by translational activation. The exact molecular mechanisms underlying this form of plasticity, downstream of translation, remain unknown; however, the identification of translationally altered genes in TRAP analysis might facilitate their discovery.

Previous studies have shown that pharmacological targeting of mTORC1 can alleviate hypersensitivity in animal models of inflammation^{18,52–55} and inhibition of mTORC1 shortly before or after nerve injury transiently alleviates pain hypersensitivity^{14,18}. These effects could be mediated through the downregulation of translation in DRG neurons, or alternatively, via inhibiting translation-independent functions of mTORC1 such as lipid biogenesis, regulation of mitochondrial functions, and autophagy, which are all implicated in neuropathic pain^{56–59}. Targeting eIF4E in the spinal cord via eIF4E-ASO does not affect DRG neurons and specifically inhibits the translational control mechanism, without affecting other functions of mTORC1.

Our study has several limitations. eIF4E-ASO was administered via the i.c.v. route to specifically target the CNS without affecting DRGs. Although we showed unaltered locomotor function using rotarod and open field tests and measured von Frey reflexive responses, we cannot completely rule out supraspinal effects of eIF4E-ASO. Notably, eIF4E-ASO is not cell-type specific; thus, it does not allow us to draw conclusions about the specific cell type(s) contributing to the pain-alleviating effects.

We used both sexes in behavioral experiments but only female mice in all other experiments. Therefore, future translational profiling studies in neuropathic pain should be extended to males. We used slightly different time points for Ribo-seq (day 63 for the late phase) and TRAP (day 60 for the late phase) experiments. Since we did not make a direct comparison between the two approaches, we believe that these differences in time points do not affect the conclusions of the

study. Finally, we used cKO mice in which Cre expression begins either during embryonic or early postnatal stages (*Tac1*^{Cre}, *Gad2*^{Cre}, and *Vglut2*^{Cre}). This may potentially lead to developmental alterations that could confound the conclusions derived from these experiments. In summary, our study provides a characterization of the translational landscape at early and late time points of neuropathic pain and in a subset of excitatory and inhibitory neurons. We identified the substantial role of spinal translation during the late stage of neuropathic pain and revealed that ASO-mediated downregulation of spinal translation provided a long-lasting alleviation of established pain hypersensitivity. These findings enhance our understanding of the cell-type- and phase-specific mechanisms underlying neuropathic pain, raising the possibility for the development of targeted therapeutics.

Materials and methods

Animals and housing conditions

C57BL/6 mice were purchased from Charles River Laboratories, Inc. (St. Constant, Quebec, Canada) at 6–7 weeks of age. On arrival at the in-house animal facility, the mice were placed in groups of 5 animals per cage with food and water ad libitum under a 12:12 h light/dark cycle (light period from 07:00–19:00 h) with ambient temperature (22 °C) and humidity maintained at 40%. *Eif4ebp1*^{fl/fl} mice⁶¹ were crossed with *Gad2*^{Cre} (The Jackson Laboratory, stock #010802) to generate *Eif4ebp1*^{fl/fl}; *Gad2*^{Cre} animals, *Pv*^{Cre} (Jackson Laboratory, stock #008069) to generate *Eif4ebp1*^{fl/fl}; *Pv*^{Cre} animals, *Tac1*^{Cre} (The Jackson Laboratory, stock #021877) to generate *Eif4ebp1*^{fl/fl}; *Tac1*^{Cre} animals, and *Vglut2*^{Cre} (The Jackson Laboratory, stock #028863) to generate *Eif4ebp1*^{fl/fl}; *Vglut2*^{Cre} animals (corresponding Cre lines were used as controls). *Tg-4EBP1mt* mice (Jackson Laboratory, stock #029735)⁴¹ were crossed with *Pv*^{Cre} (Jackson Laboratory, stock #008069) to generate *Tg-4EBP1mt*; *Pv*^{Cre} animals. *L10a-eGFP*; *Tac1*^{Cre} and *L10a-eGFP*; *Gad2*^{Cre} mice were generated by crossing *L10a-eGFP* mice⁶² with the corresponding Cre mouse lines. We also generated *Eif4ebp1*^{fl/fl}; *L10a-eGFP*; *Pv*^{Cre}, *Eif4ebp1*^{fl/fl}; *L10a-eGFP*; *Gad2*^{Cre}, and *Eif4ebp1*^{fl/fl}; *L10a-eGFP*; *Tac1*^{Cre} mice and their controls (no *Eif4ebp1*^{fl/fl}) for confirmation of 4E-BP1 deletion experiments and recording from spinal PV neurons (Fig. 5D–H [↗](#)). Sample sizes were determined based on similar previous studies in the field. Female mice were used in Ribo-seq (Fig. 1 [↗](#)), TRAP (Fig. 4 [↗](#)), FUNCAT (Fig. 3 [↗](#)) and electrophysiology experiments. Both sexes (with comparable numbers of males and females) were used in all behavioural experiments. No main effects of sex were observed; therefore, data were pooled for all reported analyses. All procedures complied with the Canadian Council on Animal Care guidelines and were approved by the McGill University's Downtown Animal Care Committee.

Spared Nerve Injury (SNI)

Mice were anesthetized under 4% isoflurane for induction and 2% for maintenance. The mice were placed on a heated (36–37°C) surgical bed during the surgical procedure. The sciatic nerve was exposed by making an incision in the upper thigh and cutting through the femoris muscle. The tibial and common peroneal branches were ligated with 7.0 silk (Covidien, S-1768K) and a 2–4-mm section of the nerve below the ligation was removed using micro self-opening scissors and forceps, leaving the sural nerve fully intact. The muscle and skin were sutured using 6.0 Vicryl (Ethicon, J489G). All sham surgeries featured incisions to the thigh and cutting of the femoris muscle, however, the sciatic nerve was left untouched and intact. Mice returned to their cage placed on a heated (36–37 °C) surface for recovery.

Western blotting

Mice were decapitated 2 weeks post-ASO injection. The lumbar section of the spinal cord, and the DRGs, were extracted and immediately homogenized in a homogenization buffer (200 mM HEPES, 50 mM NaCl, 10% Glycerol, 1% Triton X-100, 1mM EDTA, 50 mM NaF, 2 mM Na3VO4, 25 mM β-glycerophosphate, and EDTA-free complete ULTRA tablets (Roche, Indianapolis, IN) before being centrifuged at 14,000 rpm for 15 min at 4 °C to obtain a supernatant. Bradford protein assay was used to measure the protein concentration of the lysates, followed by loading 30 µg of the lysates

on a 12 % SDS-PAGE gel, and ran at a constant current (0.03 A/gel). The gel was transferred to a nitrocellulose membrane overnight on ice at 20 mV. The membrane was blocked (5% milk or BSA in TBS-T) for one hour and then incubated in primary antibodies overnight at 4 °C. Afterwards, the membrane was washed three times and incubated in HRP-conjugated secondary antibody at room temperature. The membrane was then washed three times, and an Enhanced Chemiluminescent (ECL) reagent was used to enhance the signal before visualizing it using a ChemiDoc Imaging System (Bio-Rad). Primary antibodies were eIF4E BD (Biosciences, Cat# 610270, 1:1000), and beta-tubulin (Cell Signaling, Cat# 2146S, 1:1000). Secondary antibodies were anti-Rabbit IgG – Horseradish Peroxidase antibody (GE Healthcare, Cat# NA9340 [RRID:AB_772191](#)), and sheep anti-Mouse IgG - Horseradish Peroxidase antibody (GE Healthcare, Cat # NA931; [RRID:AB_772210](#)).

Fluorescent non-canonical amino acid tagging (FUNCAT)

Female mice, aged 8–10 weeks, were fed a methionine free diet (Envigo RMS Inc., TD.110208) for one week, followed by an intraperitoneal injection of azidohomoalanine (AHA) (100 µg/g body weight, i.p., Click-IT™ AHA [L-Azidohomoalanine], Cat No. C10102, Thermo Fisher Scientific). After 3 h, mice were anesthetized and perfused transcardially with 4% PFA in phosphate-buffered saline (PBS), pH 7.4.

The L4 and L5 lumbar sections of the spinal cord were extracted and kept at 4 °C in PFA overnight. Tissue was then cut into 30-µm sections and washed three times (10-min increments) in PBST (0.2% Triton-X100 in PBS) before being blocked overnight in a solution composed of 10% normal goat serum, 0.5% Triton-X100, and 5% sucrose in PBS. Afterwards, click chemistry was performed on the sections overnight in a click buffer, consisting of 200 µM triazole ligand, 400 µM TCEP, 2 µM fluorescent Alexa Fluor 555 alkyne (Alexa Fluor™ 555 Alkyne, Cat No. A20013, Thermo Fisher Scientific), and 200 µM CuSO₄ in PBS. The sections were washed followed by immunohistochemistry (described below). Quantification (25 neurons were quantified per mouse) was performed as described below for immunohistochemistry. Anisomycin (100 mg/kg, A9789, Sigma-Aldrich) was injected intraperitoneally one hour before AHA administration.

Immunohistochemistry

Mice were anesthetized and perfused with 4% paraformaldehyde (PFA) in PBS, pH 7.4, and spinal cords were extracted and left overnight in PFA at 4 °C. Spinal cords were then transversely cut into 30-µm sections followed by three washes (10 min each) using 0.2% Triton-X100 in PBS. After washing, sections were blocked using a solution consisting of 5% normal donkey and 5% normal goat serum (NDS/NGS) and 0.2% Triton-X100 in PBS for 1 h. After blocking, tissue was incubated overnight in primary antibodies diluted in PBS. Sections were washed three times in PBS and incubated in the corresponding secondary antibody diluted in PBS for 2 h.

Primary antibodies for immunohistochemistry were: NeuN (1:1000, mouse, Millipore, MAB377), 4E-BP1 (1:200, rabbit, #2855S, Cell Signaling and Technology Laboratories), Pax2 (1:500, goat, Novusbio, AF3364) and GFP (1:1000, chicken, Abcam, ab13970). Secondary antibodies were Goat anti-rabbit Alexa Fluor 568 (1:500, Thermo Fisher Scientific, A-11011), Donkey anti-goat Alexa Fluor 647 (Thermo Fisher Scientific, A-21447), Goat anti-mouse Alexa Fluor 488 (Thermo Fisher Scientific, A-21042), and Alexa Fluor 555 alkyne (Alexa Fluor™ 555 Alkyne, Cat No. A20013, Thermo Fisher Scientific). Antibodies against eIF4E (1:500, mouse, Santa Cruz, SC-271480) and GFP (1:1000, chicken, Abcam, ab13970) were used to confirm knockdown of eIF4E in PV⁺ cells. Secondary antibodies were Goat anti-mouse Alexa Fluor 488 (Thermo Fisher Scientific, A-21042) and Goat anti-chicken Alexa Fluor 647 (Thermo Fisher Scientific, A-21449).

After three washes in PBS, the sections were mounted and imaged using a Zeiss confocal microscope (LSM 880, 63X/1.40 Oil DIC f/ELYRA objective) equipped with Argon, DPSS, and HeNe lasers. Alexa Fluor 488 was excited using the Argon laser and detected with an emission window of 507–551 nm.

Alexa Fluor 555/568 was excited using the DPSS 561 nm laser and detected with an emission window of 578–631 nm. Alexa Fluor 647 was excited using the HeNe 633 nm laser and detected with an emission window of 654–690 nm. Images were acquired using identical settings across experimental groups. Integrated density of AHA signal was measured within Pax2⁺ cells for inhibitory neurons, or Pax2⁺/NeuN⁺ cells for excitatory neurons in the superficial dorsal horn (laminae I, II and III, defined using a lamina overlay based on NeuN staining) and quantified using ImageJ on maximum intensity projection images. To confirm knockout of 4E-BP1 signal, integrated density of 4E-BP1 signal was measured in eGFP⁺ neurons using *Eif4ebp1*^{fl/fl}; *L10a*-eGFP; *Pv*^{Cre}, *Eif4ebp1*^{fl/fl}; *L10a*-eGFP; *Gad2*^{Cre}, and *Eif4ebp1*^{fl/fl}; *L10a*-eGFP; *Tac1*^{Cre} transgenic mice. Corresponding mouse lines without *Eif4ebp1*^{fl/fl} were used as controls. Background noise was subtracted from the final calculations in all experiments. For all image quantification experiments, three sections were analyzed per mouse, with three images acquired from each section (nine images per mouse). The integrated density of AHA, eIF4E, or 4E-BP1 signals in the somata of 25 neurons per mouse was quantified using ImageJ on maximum-intensity projection images (generated using the same parameters and number of optical sections), and the values were averaged per mouse and presented.

Translating ribosomal affinity purification TRAP

SNI and Sham surgeries were performed on 8–10-week-old female (*L10a*-eGFP; *Tac1*^{Cre} and *L10a*-eGFP; *Gad2*^{Cre}) mice 4 or 60 days prior to tissue extraction. TRAP-sequencing was performed as previously described^{13,62}. Mice were decapitated and the lumbar section of the spinal cord was extracted under cold and RNase free conditions and transferred to ice-cold dissection buffer (1X HBSS, 2.5 mM HEPES-NaOH [pH 7.4], 35 mM Glucose, 5 mM MgCl₂; 100 µg/mL cycloheximide and 0.2 mg/mL emetine were added just before use). Next, the extracted spinal cord tissues were homogenized in lysis buffer (20 mM HEPES-NaOH [pH 7.4], 12 mM MgCl₂ and 150 mM KCl in RNase free water; 0.5 mM DTT, 1 µL/mL Protease Inhibitors Cocktail [Roche], 100 µg/mL cycloheximide, 20 µg/mL Emetine, 40 U/mL RNasin [Promega] and 2 U/mL TURBO DNase [Invitrogen] were added immediately before use) using Minilys Personal High Power Tissue Homogenizer (Bertin Technologies) on medium speed for 10 s for a total of 8 times with 10-s intervals (incubation on ice) in a cold room at 4 °C. A post-nuclear fraction was generated from spinal cord homogenates by centrifuging at 2000 x g for 5 min at 4 °C. The supernatant was collected and NP-40 (AG Scientific) and DHPC (Avanti Polar lipids) were added at a final concentration of 1% and incubated on ice for 5 min. Afterward, post-mitochondrial fractions were generated by centrifuging samples at 15,000 x g for 10 min. A 200-µL aliquot was taken from the post-mitochondrial fraction and used as the input, and the remaining fraction was incubated overnight with protein washed G-coated Dynabeads (Invitrogen) bound to 50 µg of anti-GFP antibodies (HtzGFP-19F7 and HtzGFP-19C8 antibodies were acquired from Sloan Memorial Kettering Centre) on an end-over-end mixer. On the following day, the beads were washed four times with a high salt buffer (20 mM HEPES-NaOH [pH 7.4], 12 mM MgCl₂ and 0.35 M KCl, 1% NP-40 [AG Scientific] in RNase free water; 100 µg/mL cycloheximide and 0.5 mM DTT were added just before use) to collect the immunoprecipitated fraction (IP). After the removal of the final wash, RNA was extracted by incubating 300 µL of Trizol to the IP fraction and 600 µL to the input fraction for 10 min at room temperature. Equal amounts of ethanol were added to each of the samples before eluting the RNA using a Direct-zol RNA kit (Zymo Research) using manufacturer's protocol. RNA yields were quantified using a NanoDrop Spectrophotometer ND-1000 (NanoDrop Technologies, Inc.) and RNA quality was determined by a 2100 Bioanalyzer (Agilent Technologies).

Library generation and sequencing

Sequencing was performed on immunoprecipitated and corresponding input fractions. Three lumbar spinal cords were pooled per replicate, with a total of three replicates per group. Groups were labeled as followed: day 4 SNI Tac1 or GAD2, day 4 Sham Tac1 or GAD2, day 60 SNI Tac1 or GAD2, and day 60 Sham Tac1 or GAD2. Total RNA quality assessment, library generation, library quality check and sequencing were carried out at the Génome Québec (Montreal).

Total RNA was quantified, and its integrity was assessed on a LabChip GXII (PerkinElmer) instrument. rRNA was depleted from 70 ng of total RNA using QIAseq FastSelect (Human/Mouse/Rat 96rxns). cDNA synthesis was achieved with the NEBNext RNA First Strand Synthesis and NEBNext Ultra Directional RNA Second Strand Synthesis Modules (New England BioLabs). The remaining steps of library preparation were performed using and the NEBNext Ultra II DNA Library Prep Kit for Illumina (New England BioLabs). Adapters and PCR primers were purchased from New England BioLabs. Libraries were quantified using the KAPA Library Quantification Kits - Complete kit (Universal) (Kapa Biosystems). The average size fragment was determined using a LabChip GXII (PerkinElmer) instrument.

The libraries were normalized and pooled and then denatured in 0.05 N NaOH and neutralized using HT1 buffer. The pool was loaded at 175 pM on an Illumina NovaSeq S4 lane using Xp protocol as per the manufacturer's recommendations. The run was performed for 2 x100 cycles (paired-end mode). A phiX library was used as a control and mixed with libraries at 1% level. Base calling was performed with RTA v3.4.4. Program bcl2fastq2 v2.20 was then used to demultiplex samples and generate fastq reads. mRNA library preparation and sequencing were done at Genome Quebec.

TRAP bioinformatics and statistical analysis

Mapping and TPM quantification

Quality of FASTQ files were checked using FastQC (Babraham Bioinformatics, <https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>). Phred scores, per-base sequence, and duplication levels were analysed. Reads were soft-clipped (12 bases per read) to ignore adapters and low-quality bases during alignment using STAR v2.7.6 with the GRCm39 mouse reference genome (Gencode release M31, primary assembly)⁶³. Reads were also sorted with STAR and then deduplicated with sambamba v0.8.2⁶⁴. StringTie v2.2.1 was used to obtain Transcript Per Million (TPM) values for each gene of all samples⁶⁵. Non-coding and mitochondrial genes were removed and TPM values for coding genes were re-normalized to sum to 1 million before performing downstream analysis.

Order statistics and re-normalization of expression data

Downstream analysis of TRAP datasets was done as previously described^{62,66}. Each transcriptome sample (INPUT) had consistently expressed genes identified by calculating percentile ranks for each coding gene. We identified between 14,763 and 14,812 genes, dependent on condition, that were above the 30th percentile in each INPUT sample. Quantile normalization was performed based on the set of all coding genes. Immunoprecipitation (IP; translatome) analysis was performed on the consistently transcriptome-expressed genes. To identify consistently expressed genes in the translatome samples, the percentile ranks of TPM were calculated for each of the consistently transcriptome-expressed coding genes of each sample. We identified between 13,140–13,191 out of 14,763–14,812 genes, dependent on condition, that are consistently detected in the translatome based on whether their expression was on or above the 10th percentile in each IP sample.

Differential expression (DE) analysis

Differential expression analysis was done as previously described⁶⁷. Log₂-fold change was calculated based upon median TPM values for each transcriptome-expressed and translatome-expressed coding gene in the INPUT and IP samples respectively. Strictly standardized mean difference (SSMD) was used to reveal genes with systematically altered expression percentile ranks between SHAM and SNI mice of the same cell-type and timepoint. SSMD is the difference of means controlled by the variance of the sample measurements. SSMD measured the effect size to control for within-group variability. Differentially expressed genes were determined between Sham and SNI of the same cell-type and timepoint by calculating the Bhattacharyya distance⁶⁸. This measure is used to calculate the amount of overlap in the area under the curve of the two sample distributions (corresponding to each group). BD compares distribution of gene relative to abundance (in TPMs). The Bhattacharyya coefficient BC(Q)_i ranges between 0 (for totally non-

overlapping distributions) and 1 (for completely identical distributions) and is derived from the Bhattacharyya distance. In our analysis, we used a modified form of the Bhattacharyya coefficient that ranges between 0 (for completely identical distributions) and +1 or -1 (for totally non-overlapping distributions, sign defined by the log-fold change value). DE genes were identified if the absolute value of SSMD was higher or equal to 0.97, the absolute value of BC was higher or equal to 0.5 and fold change higher or equal to 1.33. Coding for bioinformatics analysis and data visualization was done in Python (version 3.7 with Anaconda distribution).

Behavioral pain studies

von Frey

Mice were habituated for 1 h in individual transparent Plexiglas cubicles (5 cm x 8.5 cm x 6 cm) placed on a perforated steel floor. Mice were then tested by applying calibrated nylon monofilaments perpendicular to the surface of the hind paw for three seconds. Withdrawal of the mouse's foot before the monofilament buckled was considered a positive response. The up-down method of Dixon was used to estimate the 50% withdrawal threshold (average of two measurements separated by at least 30 min)⁶⁹.

Radiant heat paw-withdrawal (Hargreaves') assay

Mice were habituated for 1 h in individual transparent Plexiglas cubicles (5 cm x 8.5 cm x 6 cm) placed on a transparent glass floor. During testing, a high-intensity light source was applied to the surface of the hind paw. Intensity was set at 20% (of the maximum) using the IITC model 390. Latency of the hind paw withdrawal was measured. The cut-off for a response was 40 s. Hind paws were measured twice separated by at least 30 min.

Mouse Grimace Scale (MGS)

The Mouse Grimace Scale was adopted from a previous publication⁷⁰. Mice were habituated in custom-made Plexiglas cubicles (5.3 × 8.5 × 3.6 cm) for 1 h. After habituation, mice were recorded (Sony Digital Camcorder HDR-PJ430V) for 1 h. One photo was chosen from every 3-min period for a total of 20 pictures. These pictures were then randomized, and the coder was blinded to the groups before they were analyzed to give each mouse their mean score. Scoring was based off five facial features (action units) including: orbital tightening, nose bulge, cheek bulge, ear position, and whisker change. Scores ranged from 0 to 2, with 0 being no evidence of the action unit present, 1 moderate evidence of the action unit and 2 obvious evidence of the action unit. The final MGS score was given to each mouse based upon averaging the intensity ratings for all five action units.

Rotarod test

Motor coordination was evaluated using the IITC Life Science Rotarod apparatus. Mice were first habituated to the task by balancing on a rotating rod set at a constant speed of 5 rpm for 1 minute. This habituation session was conducted three times, with 5-minute intervals between trials. During the test session, the rod initially rotated at 5 rpm and gradually accelerated at a rate of 0.2 rpm/s over a 5-minute period. The latency to fall was automatically recorded when the animal dropped onto the sensor located beneath the rod. Each mouse underwent three test trials, separated by 10-minute intervals. Both the time to fall (s) and the rotational speed at the moment of falling (rpm) were measured.

Open field test

General locomotor activity was assessed by allowing mice to freely explore an open field arena (44 × 44 cm) for 5 minutes. Movement was recorded and analyzed using an automated video-tracking system (Ethovision 5.0, Noldus) to determine total distance traveled and average velocity.

I.c.v. stereotaxic injection

Mice were deeply anesthetized (induced with 5% isoflurane and maintained on 2% isoflurane) and their head was secured using ear bars in a stereotaxic frame (Kopf). The head was shaved, and an incision was made to expose the skull. Aiming occurred via coordinates relative to bregma (anterior/posterior (AP): -0.5 mm, medial/lateral (ML): 1 mm, and dorsal/ventral (DV): -2.2 mm), and the skull was drilled for injection of the ASO into the lateral ventricles (i.c.v.). Five μ L of eIF4E ASO or scrambled ASO was injected using a 10- μ L Hamilton microsyringe mounted on a perfusion pump. The perfusion rate was set to 0.5 μ L/min, and the needle stayed in place for an additional 5 min to prevent leakage. The timeline of ASO administration, SNI, and behavioral testing is shown in Fig. 2 [↗](#).

eIF4E antisense oligonucleotide

eIF4E ASO targeting mouse *Elf4e* mRNA and Control scrambled ASO were developed and synthesized by Ionis Pharmaceuticals, see Supplementary Table 2 for sequences. Each ASO consists of 5 nucleotides on the 5' and 3' ends of the ASO with a 2'-O-methoxyethyl (MOE) modification, and a central 10-base DNA 'gap'. ASO were diluted in phosphate-buffered saline (PBS) to a concentration of 100 mg/ml and were delivered i.c.v. at a single dose of 100 mg/kg.

Intraspinal AAV injections

Eight-10-week-old *Pv^{Cre}* mice were injected with AAV-CAG-DIO-eGFP-eIF4E-shRNAmir or AAV-CAG-DIO-EGFP-scrambled-shRNAmir using a minimally invasive (non-laminectomy) technique. Mice were deeply anesthetized (induced with 5% isoflurane and maintained on 2% isoflurane) and steel clamps were attached to the vertebral column. An incision was made in the skin and muscle at T12-L3, removing the muscle from the space between the T13 and L1 vertebrae. A glass electrode was inserted 250 μ m into the spinal dorsal horn. A total of 500 nL of AAV-shRNA was injected over a 10-min period (David Kopf instruments, 99236B) followed by suturing of the skin using 6-0 Vicryl silk (Ethicon, J489G).

AAV9-shRNAmir cloning and preparation

The microRNA-adapted short hairpin RNAs (shRNAmir) packaged in adeno-associated virus (AAV9-CAG-DIO-eGFP-eIF4E-shRNAmir and AAV9-CAG-DIO-EGFP-scrambled-shRNAmir) were prepared by Vector Biolabs. The validated sequence targeting mouse eIF4E was: TCCAGTTGTCTTAATTTAAGTCA GTCAGTGGCCAAACTTAAATTACTAGACAACTGGACAG, and the scrambled sequence used as a control was: GAAATGTACTGCGCGTGGAGACGTTTTGGCCACTGACTGACGTCTCCACGCAGTACATTTT AG.

Electrophysiology

Tissue preparation

Six-8-week-old mice were anesthetized with 200 mg/kg tribromoethanol (Avertin, i.p.) and cardiac perfused with 4 °C NMDG-ACSF solution containing (in mM): 92 NMDG, 2.5 KCl, 1.2 NaH_2PO_4 , 30 NaHCO_3 , 20 HEPES, 25 Glucose, 5 Sodium Ascorbate, 2 thiourea, 3 Sodium Pyruvate, 10 MgSO_4 , 0.5 CaCl_2 (pH = 7.3–7.4, 300–310 mOsm). Following the cardiac perfusion, the vertebral column was rapidly removed and placed in the same oxygenated NMDG-ACSF solution described above. The vertebrae were removed with microforceps and microscissors, under a Zeiss Stemi 305 stereo microscopes, and the dorsal/ventral roots were clipped close to the dorsal root ganglia. The spinal cord of the lumbar region was carefully peeled from the dura matter and superfluous roots and glued to a 2% agar block with the dorsal side up, and then embedded in 3% low melting point agarose. Transverse parasagittal slices (250- μ m thick) were made using a Vibratome (Leica VT1200). Slices were incubated at room temperature for 30–45 min in oxygenated recovery solution containing (in mM): 92 NaCl, 2.5 KCl, 1.2 NaH_2PO_4 , 30 NaHCO_3 , 20 HEPES, 25 Glucose, 5 Sodium Ascorbate, 2 thiourea, 3 Sodium Pyruvate, 2 MgSO_4 , 2 CaCl_2 (pH = 7.3–7.4, 300–310 mOsm).

Whole-cell recordings

During recording, the spinal cord preparation was kept submerged and perfused with ACSF containing (mM): 125 NaCl, 25 NaHCO₃, 1.25 KCl, 1.25 KH₂PO₄, 1.5 MgCl₂, 1.5 CaCl₂, and 16 glucose and saturated with 95% O₂-5% CO₂, the temperature was kept constant (within ± 0.5 °C), at 30 °C. Fluorescent-labelled PV neurons at the lamina II-III in the dorsal region of the spinal cord slices were visually identified under the fluorescent microscope (Zeiss axiocam 506). To measure the intrinsic membrane properties and neuronal excitability, whole-cell recordings were performed in current-clamp mode with an increment step of 20 pA (0 – 180 pA), 1-2 second current injections. The patch pipettes were filled with intracellular solution containing (in mM): 140 K-gluconate, 2.5 MgCl₂, 10 HEPES, 2 Na₂-ATP, 0.5 Na₂-GTP and 0.5 EGTA (pH 7.3, 295–305 mOsm), and recordings were excluded if the RMP was more positive than -50 mV or series resistance was >25 M Ω .

The data were acquired with pCLAMP 11.0 (Molecular Devices) at a sampling rate of 10-20 kHz and were measured and plotted with pCLAMP 11.0 and GraphPad prism 9.0. Results are reported as mean $\pm$ SEM. Statistical analysis of the data was performed using unpaired t-test or one-way ANOVA, followed by Tukey's multiple comparison test. Statistical significance was set at $p < 0.05$.

Ribo-seq

Tissue extraction

On day 4 and day 63 post-SNI or sham surgery, mice were euthanized by isoflurane anesthesia followed by decapitation. Immediately after, the mouse was held vertically, with the rostral end facing down allowing blood to drain from the trunk for about 15–20 s. Then the mouse body was promptly placed and secured on a bed of dry ice using duct tape. The spinal column was carefully cut open to expose the spinal cord and the DRGs, which were doused with RNAlater (Invitrogen, AM7020). The lumbar DRGs were extracted from the L3, L4 and L5 vertebrae levels and the corresponding section of the spinal cord was extracted from the T12, T13, and L1 vertebrae levels. The extracted tissues were quickly placed in pre-chilled, RNase-free microcentrifuge tubes (Ambion) and snap frozen by submerging the tubes in liquid nitrogen for a few seconds. The tubes were then stored in a -80 °C freezer until tissue from all animals was collected. Tissue from 15 animals was pooled for each ribosome footprinting (Ribo-seq) replicate.

Ribosome footprinting and RNA-Seq

All consumables and solutions used were certified RNase free by the manufacturer. Bench-top, pipettes, centrifuge rotors, gel tanks, glass Dounce Homogenizers were cleaned with RNaseZap (Invitrogen, AM9780) as per manufacturer's instructions.

Tissue homogenization

Tissue homogenization, RNA and footprint extraction, and library preparation were carried out according to the protocol described by Ingolia et al.,¹⁹ with minor modifications as outlined in Uttam et al.¹²

Briefly, frozen tissue (DRGs or spinal cord, pooled from 15 animals per replicate) was homogenized in 800 μ l lysis buffer using pre-chilled 2-ml glass Dounce Homogenizers, performing 30 strokes each with pestle A followed by pestle B. The tissue lysate was collected in a microcentrifuge tube and centrifuged at 16,000 $\times g$ for 15 min at 4 °C. The total crude RNA content in the supernatant was determined using a NanoDrop1000. A fraction of tissue lysate containing 100 μ g total crude RNA was reserved for mRNA-Seq and the remaining was used for ribosome footprinting.

Nuclease footprinting

It was ensured that one complete set of replicates had the same amount of crude total RNA to start with. The RNase I treatment was carried out as described by Ingolia et al.¹⁹ for 45 min at 4 °C using 5 μ l of RNase I (Ambion, AM2295) per 250 μ g of crude total RNA and quenched by adding 20 μ l of SUPERase-In (Ambion, AM2694) per 5 μ l of RNase I used.

Recovery of ribosome-associated footprints

Ribosome-associated footprints were pelleted by ultra-centrifuging the RNase I digestion mix (final volume 540 μ L) layered on top of a 660 μ L sucrose cushion at 71,000 rpm at 4 °C in a Beckman Coulter TLA-120 rotor. The ribosomal pellet was resuspended in 600 μ L of 10 mM Tris pH 7 (prepared from 1 M Tris pH 7, Ambion) and stored at –80 °C until all samples for each tissue type were processed until this step.

RNA was purified from the resuspended ribosomal pellet using the Hot-Phenol RNA Extraction method and precipitated with isopropanol. For the Hot-Phenol RNA extraction, the resuspended ribosomal pellet was brought to a final volume of 700 μ L with 10 mM Tris pH 7 and supplemented with 40 μ L of 20% SDS followed by heated incubation at 65 °C for 1 min at 1400 rpm in a thermomixer. The heated sample-SDS mix was equally split into two microcentrifuge tubes, containing hot acidic phenol (heated to 65 °C) and incubated at 65 °C, 1400 rpm for 5 min in the thermomixer. Subsequently, the tubes were chilled on ice for 5 min and then centrifuged at 2000 $\times$ *g* for 3 min at 4 °C to obtain a top aqueous phase, which was promptly collected in a fresh tube. Seven hundred μ L of acidic phenol was added to the aqueous phase and incubated at 25 °C, 1400 rpm in a thermomixer followed by centrifugation to again obtain the top aqueous phase. The top aqueous phase was promptly collected in a fresh tube, to which 600 μ L of chloroform was added and mixed by vortexing for 1 min at room temperature. The tubes were then centrifuged to recover the top aqueous phase which was transferred in a fresh tube. The volume of the aqueous phase was estimated using a pipette and 1/9th volume of 3M sodium acetate pH 5.5 (to a final concentration of at least 0.3 M sodium acetate) and 2 μ L of 15 mg/ml Glycoblue were added. The tube was vortexed briefly and 1 volume of pre-chilled isopropanol was added followed by overnight storage at –80 °C overnight to aid precipitation. The next morning, the tubes were centrifuged at 20,000 $\times$ *g* at 4 °C for 30 min to pellet the precipitated RNA. The supernatant was discarded, and the pellet was washed with 750 μ L of ice-cold 80% ethanol, followed by centrifugation at 20,000 $\times$ *g* at 4 °C for 5 min. The supernatant was discarded, and the pellet was air dried for up to 5 min before resuspending in 21 μ L of 10 mM Tris, pH 7. One μ L of the purified RNA was used to assess the RNA concentration on NanoDrop1000.

Purification of footprint fragments and dephosphorylation

Purification of footprint fragments from the ribosome-associated footprint complex and subsequent dephosphorylation was carried out as described in steps 18 to 29 of ref.¹⁹. In step 25, we selected Rapid Gel Extraction method to extract RNA from polyacrylamide gels. The resultant RNA pellet (dephosphorylated footprints) was resuspended in 9 μ L of 10 mM Tris pH7. One μ L of the RNA was used to assess the size and concentration of the footprints obtained on the Agilent Bioanalyzer using a Small RNA Bioanalyzer Kit and manufacturer's protocol. *rRNA depletion*. Half of the obtained dephosphorylated footprint RNA was processed further for rRNA depletion using Ribo-Zero Gold rRNA Removal Kit (Human/Mouse/Rat) (Illumina, MRZG12324). For each reaction, 90 μ L of magnetic beads were transferred to a RNase-free tube and washed two times (1 min each) with an equal volume of RNase-Free water (Ambion, 4387936) using a magnetic stand and resuspended in 35 μ L of resuspension solution. To 4 μ L of dephosphorylated RNA samples, 12 μ L of RNase-free water, 2 μ L of Ribo-Zero Reaction Buffer, and 2 μ L of Ribo-Zero rRNA Removal Solution was added, mixed by pipetting and incubated at 68 °C for 10 min, followed by a 5-min incubation at room temperature. The treated RNA samples were then transferred to a 1.5-ml microcentrifuge tube containing 35 μ L of washed magnetic beads and immediately mixed by pipetting at least 10 times followed by vortexing the tube for 10 s at medium setting. The treated RNA sample and beads mix were incubated at room temperature for 5 min, after which the tubes were vortexed for 10 s at medium speed and incubated at 50 °C for 5 min. Following this incubation, the tubes were immediately placed on a magnetic stand for at least 1 min and the supernatant was collected in a fresh 1.5-ml microcentrifuge tube. The rRNA depleted samples were purified by adding 100 μ L of RNase-free water, 18 μ L of 3 M sodium acetate (Ambion), and 2 μ L of 10 mg/ml glycogen, vortexed briefly followed by an addition of 600 μ L of ice-cold isopropanol and incubated at –80 °C overnight. On the following day, the precipitated RNA was pelleted and purified as described above, and the rRNA-depleted dephosphorylated footprint sample was dissolved in 10 μ L of 10 mM Tris, pH 7.

Library preparation from rRNA-depleted dephosphorylated footprint samples

The subsequent steps of linker ligation, reverse transcription and circularization were carried out according to steps 30 to 46, and the PCR amplification and barcode addition steps were carried out as per steps 55 to 64 in the protocol by Ingolia et al.¹⁹. The resulting libraries were referred to as ‘footprint libraries’.

Library preparation for mRNA fraction

Total RNA was extracted from the tissue lysate reserved for mRNA sequencing using the hot-phenol method as described above. The extracted total RNA was submitted to Génome Québec (Montreal) for quality assessment and library preparation using the NEB mRNA stranded Library preparation service. The libraries thus generated are referred to as ‘mRNA libraries’.

Sequencing and Analysis

Both the rFP and mRNA libraries were sequenced at an aimed sequencing depth of 50 million single-ended reads per sample on the Illumina HiSeq4000 or HiSeq2500 platforms. Demultiplexed sequencing data were provided by Génome Québec (Montreal) as .bam files.

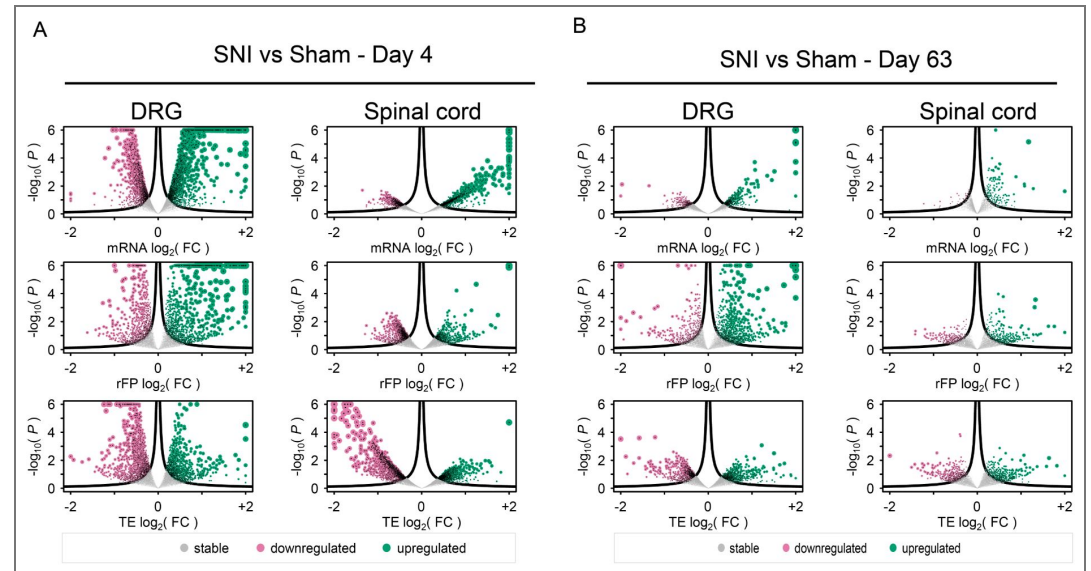
Bioinformatic Analysis for Ribosome footprinting data

The adapter sequence was identified using DREME-MEME Suite⁷¹ and trimmed from all the rFP reads using Trimmomatic⁷². Next, using Bowtie⁷², the rFP and mRNA reads were mapped to mouse reference genome mm10. Read counts for the uniquely mapped reads were generated and differentially translated genes were identified using the Xtail pipeline as described by Xiao et al.⁷³. Transcriptionally- and translationally-regulated genes were identified by Xtail using default settings. Pathway analysis was carried out using Enrichr⁷⁴, using the Kyoto Encyclopedia of Genes and Genomes (KEGG)⁷⁵ and Reactome⁷⁶ databases to identify cellular/molecular pathways associated with the differentially translated genes.

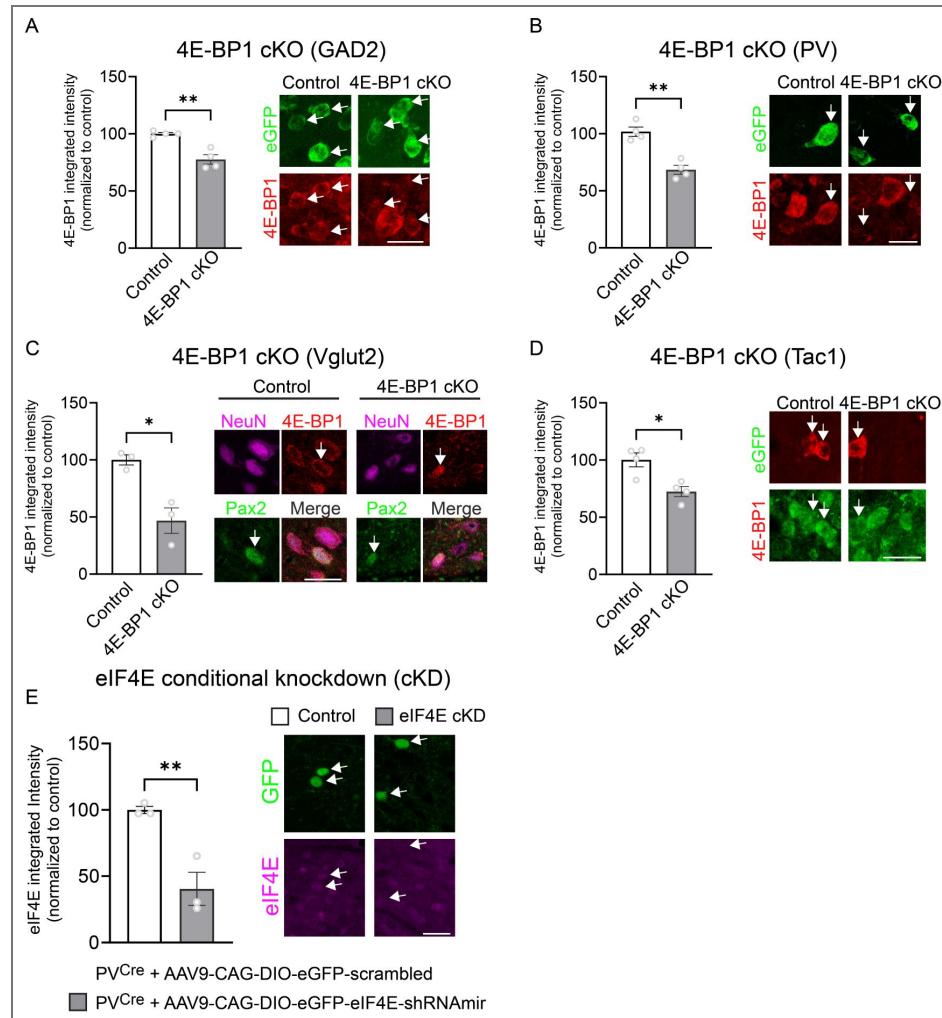
Statistical analysis

GraphPad Prism v.9 software was used to analyze data. All data are presented as means $\pm$ s.e.m. An $\alpha=0.05$ was used to determine statistical significance. Data analysis included unpaired Student’s *t*-test (two-tailed), one-way and two-way ANOVA, and repeated measures ANOVA, followed by between-group comparisons using Tukey’s *post-hoc* test, as appropriate.

Supplementary figures



Supplementary Figure 1. Volcano plots showing changes in mRNA (top), ribosome footprint (rFP, middle), and translational efficiency (TE, bottom) levels in the DRG and SC tissues at day 4 post-SNI (A) and day 63 post-SNI (B). π -values⁶⁰ calculated as $\log_2(\text{FC}) \cdot -\log_{10}(P)$, given an expression fold-change (FC; X axis) and its associated P-value (P; Y axis). Statistical significance at the $\alpha = 0.2$ level; decreased (magenta) or increased fold-change (green) expression in SNI versus sham.



Supplementary Figure 2. Confirmation of 4E-BP1 and eIF4E downregulation.

(A) Lumbar spinal cord tissue from 4E-BP1 cKO GAD2 (*Eif4ebp1*^{fl/fl}; *L10a-eGFP: Gad2*^{Cre}) and Control (*L10a-eGFP: Gad2*^{Cre}) mice was immunostained for 4E-BP1. eGFP expression indicates GAD2⁺ neurons (marked by white arrows). (B) Lumbar spinal cord tissue from 4E-BP1 cKO PV (*Eif4ebp1*^{fl/fl}; *L10a-eGFP: Pv*^{Cre}) and Control (*L10a-eGFP: Pv*^{Cre}) mice were immunostained for 4E-BP1. eGFP expression indicates PV⁺ neurons (marked by white arrows). (C) Lumbar spinal cord tissue from 4E-BP1 cKO Vglut2 (*Eif4ebp1*^{fl/fl}; *Vglut2*^{Cre}) and Control (*Pv*^{Cre}) mice was immunostained for 4E-BP1. Excitatory neurons were identified as NeuN⁺/Pax2⁻ (white arrows mark inhibitory neurons). (D) Lumbar spinal cord tissue from 4E-BP1 cKO Tac1 (*Eif4ebp1*^{fl/fl}; *L10a-eGFP: Tac1*^{Cre}) and Control (*L10a-eGFP: Tac1*^{Cre}) mice was immunostained for 4E-BP1. eGFP expression indicates Tac1⁺ neurons (marked by white arrows). (E) AAVs (AAV9-CAG-DIO-eGFP-eIF4E-shRNAmir and AAV9-CAG-DIO-eGFP-scrambled) were injected into the lumbar spinal cord of *Pv*^{Cre} mice and immunohistochemistry against eIF4E was performed 14 days later. Scale bar is 20 μ m in all images. An unpaired two-tailed *t*-test was used. Each data point represents an individual animal. Data are plotted as mean $\pm$ s.e.m. **p* < 0.05, ***p* < 0.01.

Data availability

Sequencing data generated in this study have been deposited in the Gene Expression Omnibus under the accession GSE265957.

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Additional information

Author contributions

K.C.L., S.U., J.S.M., and A.K. conceived the project, designed experiments, and supervised the research. K.C.L. performed immunohistochemistry and biochemistry analyses, surgeries, i.c.v. injections and mouse colony management. C.W., W.C., M.H. and N.B. assisted with behavioral experiments. S.U., M.A., M.P., L. D., S.M.J., Y.A., N.S. and C.G.G. performed Ribo-seq. D.T.F., N.I., K.M., and T.J.P assisted with TRAP analysis. N.G. performed electrophysiological experiments. DHT assessed locomotor functions. HTZ and BF generated ASOs. F.B. and N.S. helped with data analysis and interpretation. K.C.L., J.S.M., and A.K. wrote the manuscript, with input from all authors.

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Additional files

[Supplementary table 1.](#) 

[Supplementary table 2.](#) 

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Peer reviews

Reviewer #2 (Public review):

Summary:

This manuscript compares transcription and translation in the spinal cord during the acute and chronic phases of neuropathic pain induced by surgical nerve injury. The authors chose to focus their investigation on translation in the chronic phase due to its greater impact on gene expression in the spinal cord compared to transcription.

(1) The study is significant because the molecular mechanisms underlying chronic pain remain elusive. The role of translational regulation in the spinal cord has not been investigated in neuroplasticity and chronic pain mouse models. The manuscript is innovative and technically robust. The authors employed several cutting-edge techniques such as Rio-seq, TRAP-seq, slice electrophysiology, and viral approaches. Despite the technical complexity, the manuscript is well-written. The authors demonstrated that inhibition of eIF4E alleviates pain hypersensitivity, that de novo protein synthesis is more pronounced in inhibitory interneurons, and that manipulating mTOR-eIF4E pathways alters mechanical sensitivity and neuroplasticity.

(2) Strengths: innovation (conceptual and technical levels), data support the conclusions.

Comments on revisions:

The authors did a great job addressing my comments.

<https://doi.org/10.7554/eLife.100451.2.sa3>

Reviewer #4 (Public review):

Summary:

The significance of this study lies in its focus on translational regulation in the late phase of neuropathic pain, using both genetic and pharmacological approaches, with specific emphasis on parvalbumin-positive (PV⁺) inhibitory interneurons in the spinal cord. The authors are very responsive to all the reviewers' comments.

Strengths:

I did not review this manuscript in the first round. However, the authors have been highly responsive to the reviewers' comments and have substantially strengthened the study. They conducted new behavioral experiments that yielded informative negative results (Fig. 6A and 6B). These findings demonstrate that targeting translational control in PV neurons is

sufficient to reverse SNI-induced reductions in PV neuron excitability, but insufficient to ameliorate behavioral phenotypes. This suggests that additional cell types and pathways contribute to late-phase neuropathic pain.

Weaknesses:

Only the withdrawal threshold was measured to assess neuropathic pain. Some studies only used female mice. However, the authors appropriately discuss the study's limitations in the final two paragraphs and have added experimental details to improve clarity. Overall, the manuscript has been significantly improved.

<https://doi.org/10.7554/eLife.100451.2.sa2>

Reviewer #5 (Public review):

Summary:

This study investigates the molecular mechanisms underlying the maintenance of neuropathic pain, specifically focusing on the role of mRNA translation in the spinal cord. Using the Spared Nerve Injury (SNI) model, the authors demonstrate that while both transcription and translation are active in the early phase, the chronic phase (day 63) is uniquely characterized by a shift toward translational control. They identify spinal inhibitory neurons, particularly parvalbumin-positive interneurons, as key sites of this translational regulation.

Strengths:

Technical Rigor: The use of Ribo-seq and TRAP-seq allows for a high-resolution view of the "translatome," which more accurately reflects the functional protein output than standard mRNA-seq. Novelty: The study uncovers that reducing a single translation initiation factor (eIF4E) specifically in the CNS is sufficient to provide long-lasting relief from established chronic pain. Addressing Disinhibition: The electrophysiological evidence showing that increased translation in PV+ neurons reduces their excitability provides a clear mechanism for the "spinal disinhibition" typically seen in chronic pain.

Weaknesses:

Cell-Type Sufficiency: New experiments in the revision show that while inhibiting translation in PV+neurons restores their individual excitability, it is not sufficient on its own to reverse behavioral pain hypersensitivity. This suggests that the maintenance of chronic pain likely involves translational changes across a broader network of cell types, including other inhibitory neurons or non-neuronal cells like microglia. -This does not have to be resolved in the current study, but providing some framework to account for potential mechanisms might help the audience.

<https://doi.org/10.7554/eLife.100451.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

This study investigated the role of transcriptional and translational controls of gene expression in dorsal root ganglia and lumbar spinal cord in neuropathic pain in mice. Using ribosome profiling (Ribo-seq) and translating ribosome affinity purification (TRAP), they show changes in transcriptomic and translational gene expression at the peripheral and central levels rapidly after nerve injury. While translational changes in gene expression remained elevated for more than two months in both DRGs and the spinal cord, transcriptomic regulation was absent in the spinal cord long after the onset of neuropathy. Disrupting mRNA translation in dorsal horn neurons using antisense oligonucleotides reduced mechanical withdrawal threshold and facial expression of pain. Using fluorescent noncanonical amino acid tagging (FUNCAT), the authors further show that de novo protein expression primarily occurs in inhibitory neurons in the superficial dorsal horn after nerve injury. Accordingly, a selective increase in translational control of gene expression in spinal inhibitory neurons, or a subset of mainly inhibitory neurons expressing parvalbumin (PV), using transgenic mice, led to a decrease in the excitability of PV neurons and mechanical allodynia. In contrast, decreasing the translational control of spinal PV neurons prevented the alteration of the electrophysiological properties of the PV cells induced by nerve injury.

Strengths:

This is a well-written article that uncovers a previously unappreciated role of gene expression control in PV neurons, which seems to play an important part in the loss of inhibitory control of spinal circuits typically seen after peripheral nerve injury. The conclusions are generally well supported by the data.

Weaknesses:

The study would benefit from further clarifications in the methods section and a deeper analysis of gene expression changes in mRNA expression and ribosomal footprint observed after nerve injury.

We have improved the description of the methods and clarified the rationale underlying the presentation of gene expression changes. We have also added lists of the top differentially expressed genes at both the translational and transcriptional levels to Figure 1, and improved the description of the datasets in the Supplementary Materials.

Antisense oligonucleotides used to reduce translation by disrupting eIF4E expression were administered i.c.v. It is unknown if the authors controlled for locomotor deficits, which might add confounds in the interpretation of behavioral results. A more local route should have been preferable to avoid targeting brain regions, which could potentially affect behavior.

Thank you for raising this important point. We used i.c.v. administration to specifically target the central nervous system (CNS) without affecting the peripheral nervous system, as this is the recommended approach for selectively targeting the CNS using ASOs. Intraspinal administration of ASOs (into the spinal cord parenchyma) at an effective dose for long-term effects is not feasible. Intrathecal administration is possible but would result in exposure of the DRGs to the injected ASO and therefore would not be specific to the CNS.

To rule out potential locomotor deficits, we now subjected mice to the rotarod and open field tests to assess motor function. We found no differences between eIF4E-ASO- and control-ASO- injected mice (Fig. 2J, K).

In the revised version of the manuscript, we now better explain the rationale for i.c.v. injection. Moreover, we discuss the potential supraspinal effects of eIF4E-ASO in the

Limitations section, while also describing the lack of motor phenotypes in the rotarod/open field tests.

Only female mice were used for Ribo-Seq, TRAP, FUNCAT, and electrophysiology, but both sexes were used for behavior experiments.

Our manuscript involves various complicated techniques and analyses. Due to limited resources, we therefore opted to use only females for expensive and labor-intensive experiments, such as Ribo-Seq, TRAP, FUNCAT, and electrophysiology, while using both sexes for behavioral studies.

We now clearly acknowledge this limitation in the revised manuscript.

The conditional KO of 4E-BP1 using transgenic animals should be total in the targeted cells. However, only a partial reduction is reported in Figure S2 in GAD2, PV, Vglut2, or Tac1 cells. Again, proper methods for quantification of fluorescence in these experiments are lacking.

We apologize for the oversight; we have now updated the description of the methods for IHC signal quantification. Although genetic ablation is indeed expected to result in a complete loss of signal, in practice, previous studies employing IHC, but not Western blotting, for 4E-BP1 have also shown only a partial reduction in signal. This is likely because the 4E-BP1 antibody partially detects other epitopes. Using the same antibody, we and others have shown complete elimination of the band corresponding to 4E-BP1 in spinal cord and DRG tissue (e.g., PMID: 26678009).

The elegant knockdown of eIF4E using AAV-mediated shRNAmir shows a recovery of the electrophysiological intrinsic properties of PV neurons after injury. It is unclear if such manipulation would be sufficient to reverse mechanical allodynia in vivo.

Thank you for this concern, which was also raised by other reviewers. We have now performed two additional experiments, which revealed that suppressing the mTORC1–eIF4E axis in spinal PV neurons (using AAVs expressing eIF4E-shRNA in spinal PV neurons [Fig. 6A] and transgenic mice expressing non-phosphorylatable 4E-BP1 in PV neurons [Fig. 6B]) is not sufficient to alleviate neuropathic pain. These new findings need to be reconciled with our other results showing that eIF4E downregulation in PV neurons prevents the SNI-induced reduction in their excitability, and that ASO-mediated suppression of eIF4E, which affects all cell types, alleviates neuropathic pain.

Together, these results suggest that targeting translational control in PV neurons is sufficient to reverse SNI-induced reduction in PV neuron excitability, but is not sufficient to prevent behavioral phenotypes, which likely require changes in other cell types and/or additional pathways, as well as other alterations within PV neurons. We have now included these new results in the revised manuscript (Fig. 6A and Fig. 6B) and revised the text accordingly. These changes include toning down the role of translational control in PV neurons after SNI in driving behavioral hypersensitivity.

Reviewer #2 (Public review):

Summary:

I reviewed the manuscript titled "Translational Control in the Spinal Cord Regulates Gene Expression and Pain Hypersensitivity in the Chronic Phase of Neuropathic Pain." This manuscript compares transcription and translation in the spinal cord during the acute and chronic phases of neuropathic pain induced by surgical nerve injury. The authors chose to focus their investigation on translation in the chronic phase due to its greater impact on gene expression in the spinal cord compared to transcription.

(1) The study is significant because the molecular mechanisms underlying chronic pain remain elusive. The role of translational regulation in the spinal cord has not been investigated in neuroplasticity and chronic pain mouse models. The manuscript is innovative and technically robust. The authors employed several cutting-edge techniques such as Ribo-seq, TRAP-seq, slice electrophysiology, and viral approaches. Despite the technical complexity, the manuscript is wellwritten. The authors demonstrated that inhibition of eIF4E alleviates pain hypersensitivity, that de novo protein synthesis is more pronounced in inhibitory interneurons, and that manipulating mTOR-eIF4E pathways alters mechanical sensitivity and neuroplasticity.

Strengths:

Innovation (conceptual and technical levels), data support the conclusions.

Weakness:

Confusion about the sex of the animals. It is unclear whether eIF4E ASO affects translation and which cells. It is not determined that modulating translation in PV⁺ neurons impacts neuropathic pain behaviors.

We thank the reviewer for their thoughtful comments. In the revised version of the manuscript, we better explain that both sexes were used for behavioral experiments, whereas only females were used for Ribo-Seq, TRAP, FUNCAT, and electrophysiology experiments.

ASOs are not known to be intrinsically cell-type-specific; therefore, we do not expect differential effects on excitatory versus inhibitory neurons. We demonstrated that eIF4E-ASO reduces the levels of eIF4E, a key translation initiation factor that is rate-limiting for cap-dependent translation.

Moreover, in the revised manuscript we included two additional experiments (Fig. 6A and Fig. 6B) showing that decreased eIF4E-dependent translation in PV neurons is not sufficient to alleviate neuropathic pain, despite its effects on excitability measures. We have updated the manuscript to reflect these important new findings

Reviewer #3 (Public review):

Summary:

This study provides evidence for translational changes in inhibitory spinal dorsal horn neurons following chronic nerve injury. Gene expression changes have been widely studied in the context of pain induction and provided key insights into the adaptation of the nervous system in the early phases of chronic pain. Whereas this is interesting biologically, most patients will arrive in the clinic beyond the acute phase of their injury, thus limiting the translational relevance of these studies. Recent studies have extended this work to highlight the difference between acute and chronic pain states, potentially explaining the cascading factors leading to chronic pain, and hopefully how to prevent this in vulnerable populations. The present study suggests that translational changes within spinal inhibitory populations could underlie long-term chronic pain, leading to decreased inhibition and heightened pain thresholds.

Strengths:

The approaches used and the broad outcomes of the manuscript are interesting and could be an exciting development in the field. The authors are using approaches more common in molecular biology and extending these into neuroscientific research, getting into the detail of how pathology could impact gene expression differentially across the

course of an injury. This could open up new areas of research to selectively target not only defined populations but additionally help alleviate pain symptoms once an injury has already reached the maintenance phase. There is an opportunity to delve into what must be a very large data set and learn more about what genes are differentially translated and how this could affect circuit function.

Weaknesses:

Whereas the authors approach a key question in pain chronicity, the manuscript falls a little short of providing any conclusive data. The manuscript was in some areas very difficult to follow. Terminology was not always consistent or clear, and the flow of the manuscript could use some attention to highlight key areas. Whereas the overall message is clear in the summary, this would not necessarily be the case when reading the manuscript alone.

To improve the clarity and flow of the manuscript, we made changes to the text, including the addition of intermediate summaries and further explanations of terms and experiments.

The study claims to show that translational control mechanisms in the spinal cord play a role in mediating neuropathic pain hypersensitivity, but the studies presented do not fully support this statement. The authors instead provide some correlation between translation and behavioural reflex excitability (namely vfh and Hargreaves).

It is difficult to fully interpret the work, as there are a number of inconsistencies, namely the range of timings pre- and post-injury, lack of controls for manipulations, the use of shmiRNA versus lineage deletions, and lack of detailed somatosensory testing. It is not completely clear how this work could be translatable as is, without a deeper understanding of how translational control affects circuit function and whether all of this is necessarily bad for the system, or whether this is a positive homeostatic adaptation to the hyperexcitability of the circuit following injury.

A large portion of the work is focussed on showing an inhibitory-selective change in translation following chronic nerve injury. The evidence for this is however lacking. Statistics to show that translational effects are restricted to inhibitory subpopulations are inadequate. The author's choice of transgenic lines is not clear and seems to rely on availability rather than hypothesis.

Although we agree with some of the criticism, we have reservations regarding other points raised by the reviewer. To address several of the concerns, we added new experiments (Fig. 2J, 2K, 6A, and 6B). We also made changes to the text to improve readability and to better explain the rationale for the study and our focus on inhibitory neurons.

For example, we clarify that we do not state that changes in mRNA translation in the spinal cord during the chronic phase of neuropathic pain occur exclusively in inhibitory neurons. Although we observe changes in general protein synthesis, assessed using FUNCAT, in inhibitory but not excitatory neurons after SNI, alterations in the translation of specific transcripts, assessed using the TRAP approach, are observed in both excitatory and inhibitory neurons.

The second part of the paper focuses on inhibitory neurons because these neurons demonstrate larger translational changes. We now clearly indicate that alterations in excitatory neurons are also likely important during the chronic phase of SNI. This conclusion is further supported by newly added results (Fig. 6A and Fig. 6B), showing that targeting eIF4E-dependent translation in spinal PV neurons using two different approaches is not sufficient to reverse pain hypersensitivity.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Analysis of gene expression in Figure 1 lacks clarity, and the data do not effectively guide the reader toward their intended purpose. A list of the most dysregulated genes at the transcriptional level, the translational level, or both, would help the reader fully appreciate the outcome of this analysis. Similarly, what is the message conveyed by Figures 4 D-G?

As requested, we have now included the top 10 upregulated and top 10 downregulated genes at both the translational and transcriptional levels in Figure 1. We also expanded the main text and figure legends to clarify that Supplementary Figure 1 includes volcano plots for all conditions, and that Supplementary Table 1 contains the complete datasets. In addition, we expanded the figure legends to explain the organization of the data in Supplementary Table 1. Finally, we provide pathway analyses of translationally regulated genes in the spinal cord, as this condition is the primary focus of the study.

Figure 4D–G shows the top 15 translationally upregulated and downregulated genes in inhibitory neurons at days 4 (D) and 60 (E), and in Tac1⁺ excitatory neurons at days 4 (F) and 60 (G) (four conditions in total) after SNI. These panels convey that translational regulation of specific transcripts occurs in both inhibitory and excitatory neurons. Panel 4H further demonstrates that, although translational changes are observed in both neuronal populations, a greater number of genes are altered in inhibitory neurons. We have improved the readability and flow of this section to better convey this message.

Details about how AHA was quantified in Figure 3 are missing. It is unclear how and where the cells were selected for quantification. Objective criteria for expression/no expression of AHA in the cells are not indicated. Additionally, the signal seems to have somehow been normalized over images from the contralateral side. It is difficult to understand what the bar graphs actually represent in panel C. One would interpret them as percentages of excitatory/inhibitory cells expressing AHA.

We apologize for the lack of clarity. We have now expanded the description of the analyses in the figure legend and in the Methods to better explain the results shown in Fig. 3. The imaged cells were selected based on specific criteria, such as lamina location and cell type. In panel C (the anisomycin experiment), values were normalized to the control group. In all other panels, no normalization was applied, and the values represent the AHA integrated density on maximum intensity projection images (averaged per mouse). We also describe the number of sections and cells per mouse, as well as other technical details, as requested.

In addition, a few minor changes should be made:

(1) Rephrase Introduction: "Peripheral nerve injury can cause neuropathic pain, a chronic pain condition [...]." Neuropathic pain is not necessarily chronic.

This sentence was reworded to read "Peripheral nerve injury may result in neuropathic pain, a debilitating condition with limited effective treatment options".

(2) Host species for secondary anti-mouse antibodies are provided but not for the anti-rabbit (donkey?). Also, check for consistency in the methods section. The method mentions P21 two secondary antibodies and an apparent third antibody named "anti-HRP-conjugated antibody." Please provide information about this antibody, or remove it.

Thank you for flagging it, the inadvertent repetition of "anti-HRP-conjugated antibody" was removed.

(3) Provide primary antibody hosts on page 22.

The hosts of all primary and secondary antibodies were now provided.

| (4) Define PBST on page 21 and PBS-T on page 22.

We defined PBST in the revised manuscript (0.2% Triton-X100 in PBS).

| (5) Specify the filter sets used for fluorescent microscopy.

We specified the filter sets used for fluorescent microscopy.

| (6) Change the legend to 50% withdrawal threshold for vF behavior tests.

We addressed this by making the requested change in all relevant legends.

Reviewer #2 (Recommendations for the authors):

Major:

(1) The authors need to show that eIF4E ASO (Figure 2) reduces translation in both inhibitory and excitatory neurons.

ASOs are not intrinsically cell-type specific, as they do not contain promoters or regulatory elements and act wherever they enter cells and engage RNase H1. However, differences in ASO effects across cell types can arise from variability in uptake, intracellular trafficking, RNase H activity, or target mRNA expression levels.

In our study, we used eIF4E-ASO as a general approach to demonstrate that eIF4E-dependent translation contributes to SNI-induced hypersensitivity, particularly at the chronic phase. We show a marked reduction in eIF4E levels in the spinal cord of eIF4E-ASO-injected mice compared with controls. We do not claim that the effects of eIF4E-ASO are mediated by a specific cell type; rather, they may involve excitatory neurons, inhibitory neurons, and non-neuronal cells, such as microglia and astrocytes, among others.

Notably, while eIF4E can promote general translation during development, in adult mice it predominantly regulates cap-dependent translation of specific mRNAs without having a major effect on overall protein synthesis. In our case, the partial reduction in eIF4E is unlikely to substantially affect general translation, as assessed by AHA incorporation, and would instead require TRAP or Ribo-Seq to detect transcript-specific translational changes. We now better explain the rationale for the eIF4E-ASO experiment and clearly state that the effects observed cannot be attributed to a specific cell type.

In addition, our new results showing that inhibition of eIF4E-dependent translation in PV neurons is not sufficient to alleviate SNI-induced mechanical hypersensitivity suggest that translational changes in other neuronal and/or non-neuronal cell types contribute to hypersensitivity. This important point is now more clearly explained in the revised manuscript, and the role of PV neurons is toned down throughout the paper.

| (2) In Figure 5, it is necessary to show the effect of eIF4E-shRNA in PV+ neurons on neuropathic behaviors (von Frey and MGS).

To address this important concern, we performed two new experiments, both of which showed that inhibiting the mTORC1–eIF4E axis in parvalbumin neurons is not sufficient to alleviate neuropathic pain. First, we injected PV-Cre mice with AAV-eIF4E-shRNAmir and a scrambled control. We found that downregulating eIF4E in spinal PV neurons has no effect on SNI-induced mechanical hypersensitivity. We used a second, complementary approach to validate this finding. Specifically, we generated transgenic mice in which a non-phosphorylatable form of 4E-BP1 is expressed in PV neurons. Because non-phosphorylatable

4E-BP1 acts as a translational suppressor of eIF4E, this approach is functionally similar to eIF4E deletion.

Altogether, our findings indicate that cell-type–non-specific suppression of eIF4E using ASOs is sufficient to alleviate neuropathic pain, particularly at the chronic phase. In contrast, while activation of eIF4E-dependent translation in PV neurons (via 4E-BP1 deletion) induces pain hypersensitivity, suppression of eIF4E-dependent translation in PV neurons inhibits SNI-induced decrease in PV neuron excitability but does not alleviate pain hypersensitivity. Thus, increased eIF4E-dependent translation in PV neurons is sufficient to induce pain hypersensitivity, but targeting this pathway in PV neurons alone is not sufficient to reverse neuropathic pain.

Potential explanations for these findings include: (1) the presence of other important mechanisms in PV neurons (e.g., changes in synaptic transmission) that are translation independent; (2) the insufficiency of correcting reduced PV neuron excitability to alleviate hypersensitivity; and (3) an essential role for mRNA translation in other neuronal and/or non-neuronal cell types in neuropathic pain. We have updated the manuscript to include these potential explanations in the Discussion section.

Moderate:

(1) In Figure 2, MGS should be performed at earlier time points as well.

We performed MGS when von Frey testing, which is less noisy and less labor intensive in our hands, suggested altered phenotypes.

(2) In Figure 4B, the gene markers are different in Gad2+ and Tac1+ cells. Please show the 12 markers for both cell types.

We now better explain the selection of the markers.

(3) In Figure 5, MGS should be performed to test if the effect is limited to mechanical sensation/reactivity or extends to nociception. Additionally, do these mice exhibit altered locomotion and grip strength?

As described above, we added experiments involving downregulation of eIF4E and expression of a mutant non-phosphorylatable 4E-BP1 in PV neurons. We performed von Frey testing, which showed no effect of suppressing the mTORC1–eIF4E axis on mechanical hypersensitivity under these conditions. Given these negative results, we did not proceed with mouse grimace scale (MGS) analysis.

(4) In Figure S2E, the reduction of eIF4E does not appear to be specific to GFP+ cells.

We now replaced the representative images in this Figure.

(5) Can chronic neuropathic pain be reduced by enhancing 4E-BP1 specifically in PV+ neurons?

We added the experiment proposed by the reviewer in Fig. 6B. We found that enhancing 4E-BP1 activity, by expressing a non-phosphorylatable form of 4E-BP1 in PV neurons, is not sufficient to alleviate neuropathic pain hypersensitivity.

(6) Why did the authors not use PainFace for the MGS?

We began using manual, blinded MGS scoring, as originally described by Mogil and colleagues in 2010 (PMID: 20453868), for this project before PainFace became available around 2019 (e.g., Tuttle and Zylka) and in later versions (e.g., PMID: 39024163). For consistency, we therefore continued using the same approach throughout the experiments.

(7) In Figures 2A-C, the labeling of the bar graphs seems incorrect: is it 4E-BP1 or eIF4E immunoreactivity?

Thank you very much for noticing this; we have corrected the mistake.

(8) In Figure 1, present the data by sex.

We performed sequencing analyses only in females. This decision was based on the large number of mice and experimental conditions required for both Ribo-Seq (n = 15 mice per replicate, 3 replicates per condition, and 2 time points for SNI/Sham, ~180 mice total) and TRAP (n = 3 mice per replicate, 3 replicates per condition, 2 time points, and 2 genotypes [Tac1 and GAD2] for SNI/Sham), as well as the high cost of sequencing. Behavioral experiments were performed in both sexes. This information is clearly indicated in the Methods section, and we have now also included it in the Limitations section of the paper.

(9) While the methods state that all behavioral testing was done with equal numbers of male and female mice, it seems that several experiments were done only in females. In the absence of a strong justification, all experiments should be conducted in both sexes.

As explained above, due to the very large number of mice required for some experiments and the high cost of sample processing and sequencing, only behavioral experiments were performed in both sexes. We now clearly describe the sex of the animals used in each experiment in the figure legends.

Minor:

(1) In Figure 3, the legend is confusing and lacks labels.

We expanded the Fig. 3 legends and added labels, as requested.

Reviewer #3 (Recommendations for the authors):

Overall, the manuscript needs to be made clearer and more specific. As it stands, the logic and flow are difficult to follow. Figure legends are not always indicative of the figure and are inconsistent.

Regarding timelines:

The logic of the different timelines is not clear. Either explain why different times post-injury were chosen between experiments or keep them consistent. It seems a key message here is that the timing is important. It therefore follows that the authors should be strict about this in their own experiments. Figure 1: 4 and 63 days. Figure 2: Day 3 and weeks 8 and 12. Figure 3: Days 4 and 60. Figure 4: Days 4 and 60. Figure 5: 6 weeks. Figure S1: 4 and 60. Clarifying why these timings were used in each case and showing at the transcript level that these are most appropriate would be needed.

We thank the reviewer for carefully reviewing our manuscript. We focused on early versus late time points. For the sequencing experiments, we performed Ribo-seq at day 4 for the early time point and day 63 for the late time point, whereas TRAP analyses (and FUNCAT) were performed at day 4 for the early time point and day 60 for the late time point. These differences (day 60 versus day 63) were due to logistical issues related to sample collection. In our view, there are no major biological differences between day 60 and day 63 for the late time points, particularly because we do not perform direct comparisons across different experiments.

In other experiments, we used several time points (e.g., day 3, as well as 6, 8, and 12 weeks) either to follow the development of phenotypes or based on previous publications regarding

the timing of specific effects. We now acknowledge the potential limitation of using slightly different time points in the Limitations section of the paper.

Regarding the use of inhibitory and excitatory markers: The comparisons they made between subpopulations seem a little random- for one, the number of Tac1 positive cells in the dorsal horn is not equal to that of PV, and so the comparison seems inappropriate.

The number of cells from each subpopulation should not affect the number of DEGs. Because these analyses were performed on bulk mRNA rather than at the single-cell level, the comparisons are made between SNI and control groups within each subpopulation. Thus, the number of differentially translated genes is determined per cell type, not per individual cell.

The lack of any semblance of variability or statistics with regard to gene changes makes it difficult to assess whether these comparisons were justified experimentally. Pax2 is a developmentally regulated transcription factor, with reduced levels in the adult. Using Pax2- NeuN+ to label excitatory interneurons is therefore not appropriate for comparison. A more appropriate comparison would be to use vGluT2 and GAD67. Similarly, the use of the GAD2Cre seems a poor choice. This is a restricted population of interneurons that have been suggested to have specific roles in presynaptic inhibition. If the authors were interested in this subpopulation for that reason, then they should state so.

Pax2 is commonly used as a marker of inhibitory neurons in the spinal cord (e.g. PMID: 36323322) as in the adult dorsal horn, Pax2 protein remains expressed in nearly all inhibitory neurons, including both GABAergic (GAD65/67⁺) and glycinergic (GlyT2⁺) neurons. VGLUT2 marks terminals of IB4-binding peripheral sensory neurons as well as those of spinal cord excitatory interneurons in lamina II of the dorsal horn, complicating the analyses. We attempted using Lmx1b for excitatory neurons (Pax2 for inhibitory and Lmx1b for excitatory) but could not obtain specific and robust signal using different commercial antibodies (we have no access to non-commercial Pax2 antibody).

Regarding Cre lines, Gad2-Cre has been extensively used to target GABAergic neurons in the spinal cord. Although it is not expressed in purely glycinergic neurons, it is expressed in GABAergic and mixed GABA/glycine interneurons. Gad2-Cre is more restricted to superficial dorsal laminae I–III, which are relevant to pain processing, versus Gad1-Cre, which may also capture low-level GABAergic neurons in deep laminae and ventral horn inhibitory neurons. Moreover, there are also differences in the developmental profile, whereas Gad1-Cre is expressed earlier at embryonic stages during inhibitory neuron development, GAD2 is expressed later, in post-mitotic and mature inhibitory neurons. Because of these considerations (higher specificity to dorsal horn and later developmental expression), we used Gad2-Cre mouse line in our experiments.

Regarding cKO experiments:

It is unclear whether the deletion of Eif4ebp (which is not "ablation" as stated in the manuscript) has had any effect on the PV/GAD2 cells themselves seeing as this deletion would be a lineage deletion. One would imagine that altering transcription in such a population from early development would affect a host of neuronal and circuit properties, such as connectivity, dendritic branching, etc. The authors should show that the circuit properties were not broadly changed, not least as PV is expressed throughout the nervous system and in muscles. This could in itself explain the hypersensitivity described in their results. Experimenters should repeat the AAV shRNA_{mir} experiments in non-injured animals, and not just control animals with the scrambled sh.

We agree with the concerns related to potential developmental effects. Although it is nearly impossible to reliably and comprehensively demonstrate that circuit properties were not

altered in our cKO mice, our manuscript presents several lines of evidence supporting a role for translational control in specific cell types in the regulation of gene expression and nociception independent of developmental effects. First, our translational gene expression analyses were performed in adult WT mice and reflect SNI-induced changes in gene expression at the translational level, assessed using complementary approaches. In addition, the effects of eIF4E ASO delivered to adult animals support a role for translational control in the regulation of SNI-induced pain hypersensitivity at later stages.

Moreover, downregulation of eIF4E in PV neurons using an AAV-based approach in adult mice affects their SNI-induced excitability, further supporting a role for translational mechanisms in regulating PV neuron plasticity after peripheral nerve injury in adulthood. To acknowledge the potential developmental effects associated with 4E-BP1 deletion using Tac1-Cre, Gad2-Cre, and PV-Cre mouse lines (with PV-Cre beginning expression postnatally), we have included an explicit limitation statement in the Discussion of the revised manuscript.

We also thank the reviewer for highlighting the distinction between deletion and ablation, and we have corrected this terminology in the revised manuscript.

Regarding pain:

A large sticking point within the study is the lack of clarity of the populations they are targeting. Many of the populations mentioned are not expressed solely in the dorsal somatosensory horn and instead are also expressed in the ventral motor horn. This is particularly important with regard to the sensory tests they are performing, which rely on reflex responses. It seems these results, although interesting, are not proof of a pain effect, but rather showing changes in vfh-behaviour. To show this is a pain-specific event, and not just correlative or reflexive, the authors should perform further behavioural tests beyond vfh, Hargreaves, and the grimace scale, such as low threshold touch, rotarod, etc. How much of this effect is due to changes in reflex excitability? Would the authors expect similar results for all neuropathic models but not for chronic inflammatory states for example? Western Blot analysis at the moment is for the whole cord, which could imply changes in the ventral or intermediate horn, it could help strengthen the study to show that these changes are selective to the dorsal cord.

We have now added a new experiment showing that eIF4E-ASO has no effect on motor function in the rotarod and open field tests (Fig. 2J, K). In addition, the eIF4E-ASO experiment included in the original submission reflects supraspinal behavior, as assessed by MGS. Overall, our study includes numerous experiments and datasets. While we agree with some of the reviewer's concerns, the extensive additional work requested, including additional neuropathic and inflammatory pain models, further assays of supraspinal behavior, Western blot analyses restricted to the dorsal horn, additional Cre lines and markers, and other analyses, is not feasible within the scope of the current manuscript.

Notably, in the revised manuscript, we have added new experiments (Fig. 2J, 2K, 6A, 6B) that we believe address the most critical concerns raised by the reviewers, and we have revised the text to more clearly acknowledge the limitations of the study.

Regarding patch clamp studies:

An increase in rheobase alone in the PV cells would not in itself account for the changes seen in behaviour, seeing as the authors are suggesting this is a selective effect for von Frey and not radiant heat, for example. The authors should therefore show a change in mechanically-evoked firing of PV/GAD2 cells either by dorsal root stimulation in slice, or by cfos or equivalent marker of activation following sensory stimulation. The title of this figure is also misleading- it is not clear how there is any proof of promotion of plasticity in the experiments shown.

In the original submission, in addition to an increase in rheobase, we also demonstrated decreased spiking activity in response to a range of stimulating currents (Fig. 4). We agree that assessing mechanically evoked responses of PV neurons would be informative; however, such studies are beyond the scope of the current manuscript.

To address the final concern, we modified the title of Fig. 5 and the related text. Moreover, the newly added data showing that inhibition of translation in PV neurons does not alleviate SNI-induced hypersensitivity prompted us to tone down, throughout the manuscript, the link between translational changes in PV neurons and pain hypersensitivity.

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